

# SCI-MAP

## Ordinary Mapmaking and Observation Coaddition

### Formal Scientific/Engineering Contract

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## Physical model and contract at a glance

MAP consumes the realized MAP-facing output of PTC; it never bypasses PTC to consume CAL output directly. ALIGN/AST supplies the coordinate facts and MAP supplies the target grid and declared projection. SCI-MAP applies one fixed, positive-coefficient linear averaging operator to the signal and to every declared linear companion. A coadd then applies a second positive-coefficient operator to atomically admitted, compatible observation bundles on the same grid. All scientific statements are conditional on the complete admitted calibration, alignment, eligibility, coefficient, response, geometry, and covariance state.

| Facet | Compact authority                                                                                                                                                                                                                                                                                                                                                                                         |
|-------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Input | Realized SCI-PTC v0.1/r0.5 calibrated- <i>x</i> -derived nonpolarimetric total-intensity-equivalent product in <b>mJy/beam</b> , with exact originating nominal-beam identity, occurrence/generation, retention, causes, and response state; exact AST RTC-grid coordinates for the same stable sample and parent; and an explicitly selected, completely handed-off admitted PTC MAP coefficient family. |

|          |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Output   | A complete immutable observation-map bundle and, when requested and admissible, a complete centered-integer common-grid observation coadd bundle.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       |
| Equation | On admitted membership, $N_p = \sum_i a_{pi} z_i$ and $Q_p = \sum_i a_{pi}$ . The normalized scientific vector contains exactly the rows authorized by the effective support policy, with $\hat{m}_p = N_p/Q_p$ there; response and covariance use the same explicit row selection. Coadd repeats this restricted structure over complete observations.                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Source   | Frozen SCI-PTC v0.1/r0.5 digest 8357961a49272adc40e27a8aa9e760e0d01ff2419ae2c88a62c0f93c9f959e66; SCI-CAL v0.1 science r0.5/engineering r0.4 digest 413426f49edf1249f751a05bb8c6e9fd907b11e8da0530fe2da39814885efb22; SCI-AST v0.1/r0.3 manifest digest b54b6013750540f28aad02339a60bf36078980dc53b132beab73069d66ef3601; SCI-VAL Core v0.1/r0.3 digest 2fc3b3ad329fe3035d442b43d1e564a74fc86ab49f85f56e87322d8553fad9a6; SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4; the admitted uniform Registry/family/profile and uniform VAL binding; and the exact inherited MAP/AST profiles and boundaries. The exact r0.4 successor source and paired VAL binding records are supplied as CANDIDATE records; scientific-owner acceptance, canonical activation, and freeze remain pending. |
| Status   | Contract author draft. Algebraic predictions are preregistered below; none has been executed here. All stronger claim layers remain unassessed.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         |

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# 1 Scientific boundary and estimand

The ordinary observation map estimates the coefficient-weighted average of the admitted realized PTC sample quantity in each declared map pixel. Its meaning is inherited through the required CAL-to-PTC route, from ALIGN/AST's coordinate relation, and from PTC's transformed signal, product identity, availability, producer-local validity, and MAP-facing coefficient. SCI-MAP does not repair or reinterpret those facts. VAL may register and evaluate a named rule, but it is not the author of producer or MAP policy.

Let  $i$  bind observation, detector occurrence and UID, stable RTC output sample  $n$ , exact PTC product/application generation, segment, and array/network/group; time is an attribute and storage positions are locators. Here  $z_i = Z_i^{\text{PTC}}$  is the realized PTC transformed signal and  $\gamma_i$  is its exact PTC-owner-selected MAP-facing coefficient. The paired  $x/r$  symbols remain KID readout coordinates. The effective coefficient is  $a_{pi} = G_{pi}\gamma_i$ . The admitted `SCI-PTC:uniform_constant@draft-0.1` supplies exact dimensionless  $\gamma_i = 1$ , but neither Registry presence nor MAP permission selects or realizes it. Use requires explicit selection, complete atomic PTC publication, exact target/payload/occurrence association, a requested, applicable, eligible, realized `SCI-PTC:uniform_coefficient_handoff@draft-0.1` decision, independent MAP admission, and MAP finite-positive classification. An absent selection or incomplete or mismatched handoff makes the affected route unavailable; there is no unity, alternate-family, loading, sensitivity, scatter, or precision fallback.

The fixed contribution set  $\mathcal{C}_p$  factors transformed-signal availability, PTC retention, coefficient availability, coefficient/QC, `SCI-MAP:map_upstream_admission@2`, AST validity, finite signal, finite positive coefficient, admitted projection/boundary, and MAP-local companion gates. Each retained typed state  $\sigma_g$  is projected to a Boolean membership predicate  $b_g$ ; T/F/U/C and VAL's four axes are never conjoined directly. Evaluation follows the structural, safe numerical-classification, coordinate/product, membership, and accumulation stages. Only structurally admitted payloads are read for MAP classification, and all required gates pass before payload arithmetic. Every source state, immutable reason, cause, provenance, and failure scope is retained. Let  $\Pi_{\text{eff}}$  be the recorded effective product support policy and  $\mathcal{S}_{\text{out}}(\Pi_{\text{eff}})$  the exact set of map rows for which every support predicate that policy designates applicable passes. This is a row domain, not a numerical mask. The frozen ordinary result is then reused directly:

$$a_{pi} = G_{pi} \gamma_i, \quad (1)$$

$$\mathcal{G} = \{Z_{\text{avail}}, \text{PTC}_{\text{ret}}, \gamma_{\text{avail}}, \gamma_{\text{QC}}, \text{MAP}_{\text{adm}}, \text{AST}_{\text{coord}}, Z_{\text{finite}}, \gamma_{>0}, G_{\text{place}}, \text{MAP}_{\text{local}}\}, \quad (2)$$

$$b_g(i, p) = \text{pass}_g(\sigma_g(i, p)) \in \{0, 1\}, \quad g \in \mathcal{G}, \quad (3)$$

$$\text{pass}_{\gamma, \text{QC}}(i) = \mathbf{1} \begin{bmatrix} \text{request}_{\gamma, \text{QC}}(i) = \text{requested}, \\ \text{applicability}_{\gamma, \text{QC}}(i) = \text{applicable}, \\ \text{eligibility}_{\gamma, \text{QC}}(i) = \text{eligible}, \\ \text{realization}_{\gamma, \text{QC}}(i) = \text{realized} \end{bmatrix}, \quad (4)$$

$$\mathcal{C}_p = \left\{ i : \bigwedge_{g \in \mathcal{G}} b_g(i, p) = 1 \right\}, \quad (5)$$

$$\mathcal{S}_{\text{out}}(\Pi_{\text{eff}}) = \{p : \text{authorize}_{\Pi_{\text{eff}}}(p) = 1\}, \quad (6)$$

$$N_p = \sum_{i \in \mathcal{C}_p} a_{pi} z_i, \quad Q_p = \sum_{i \in \mathcal{C}_p} a_{pi}, \quad (7)$$

$$K_p = \sum_{i \in \mathcal{C}_p} a_{pi} k_i, \quad (8)$$

$$\hat{m}_p = N_p / Q_p, \quad \hat{k}_p = K_p / Q_p, \quad p \in \mathcal{S}_{\text{out}}, \quad (9)$$

A finite exact-zero signal numerator  $N_p = 0$  is a valid measured outcome when  $Q_p > 0$ , the row is support-authorized, and every other required gate passes; it gives  $\hat{m}_p = 0$ . This is distinct from  $Q_p = 0$ , which authorizes neither division nor a supported scientific row, and from a fabricated zero substituted for unavailable support.

The ten projections in Equation (3) are exact:

| Pass predicate                        | Owner and object                         | Retained source codomain                                                                       | Sole passing state                                                        | Nonpass and scope                                                                                                                    |
|---------------------------------------|------------------------------------------|------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| $b_{Z_{\text{avail}}}(i)$             | PTC; transformed occurrence              | signal availability with cause                                                                 | exact <b>available</b> $Z_i^{\text{PTC}}$                                 | unavailable, invalid, not requested, or conflict; occurrence/route exclusion                                                         |
| $b_{\text{PTC}_{\text{ret}}}(i)$      | PTC policy and VAL; occurrence           | named request, applicability, eligibility, and realization fields                              | requested, applicable, <b>eligible</b> , realized                         | every other field combination; occurrence/route scope                                                                                |
| $b_{\gamma_{\text{avail}}}(i)$        | PTC; coefficient slot/generation         | family/generation, index/broadcast, product compatibility, readable payload, typed state/cause | exact compatible typed payload available; no finiteness claim             | missing/unreadable, unavailable, conflict, or generation mismatch; route exclusion                                                   |
| $\text{pass}_{\gamma_{\text{QC}}}(i)$ | PTC policy owner; coefficient occurrence | named request, applicability, eligibility, and realization fields plus causes                  | requested, applicable, eligible, and realized under the exact PTC profile | any other named-field combination is nonpassing; structural source-binding failure only when separately present; never MAP admission |
| $b_{\text{MAP}_{\text{adm}}}(i)$      | MAP policy and VAL; occurrence           | named request, applicability, eligibility, and realization fields                              | requested, applicable, <b>eligible</b> , realized                         | every other field combination; occurrence exclusion or source-binding block                                                          |
| $b_{\text{AST}_{\text{coord}}}(i)$    | AST; same- $n$ coordinate parent         | availability, validity, frame, parent, cause                                                   | exact coordinate available and valid on matching parent                   | invalid, unavailable, conflict, or parent mismatch; occurrence exclusion                                                             |
| $b_{Z_{\text{finite}}}(i)$            | MAP numerical gate; $z_i$                | finite classification                                                                          | finite                                                                    | NaN, Inf, or unrepresentable; occurrence/product scope                                                                               |
| $b_{\gamma_{>0}}(i)$                  | MAP numerical gate; $\gamma_i$           | positive; zero; negative; non-finite; unrepresentable class                                    | finite and strictly positive                                              | zero noncontributing; negative, non-finite, or unrepresentable invalid; occurrence scope                                             |
| $b_{G_{\text{place}}}(i, p)$          | MAP; occurrence-pixel relation           | one-hot placement/boundary state                                                               | unique in-grid half-open containing pixel                                 | invalid coordinate, outer-upper boundary, or out of grid; pixel nonmembership                                                        |
| $b_{\text{MAP}_{\text{local}}}(i, p)$ | MAP; product row and role                | local and companion decisions                                                                  | every applicable local predicate passes                                   | false, unknown, conflict, unavailable, or missing companion; declared scope                                                          |

Boolean zero here means estimator nonmembership only. It is never multiplied into  $z_i$ , and it never erases the retained typed record or cause.

The supported division is not a recipe for filling unsupported pixels. Pre-normalized accumulators and representation-specific storage slots may exist outside  $\mathcal{S}_{\text{out}}$ , but the normalized scientific vector has no row there. A dense zero, sentinel, or coincidentally finite payload is not a semantic substitute and is not measured zero sky. Likewise,  $Q_p$  is a normalization fact. Its dimensions or historical name do not make it precision.

The active role is a calibrated- $x$ -derived nonpolarimetric total-intensity-equivalent quantity in **mJy/beam**. The exact originating fixed-nominal-beam/template and CAL/PTC lineage are part of the quantity identity. Neither the unit nor a STOKES token promotes it to Stokes I. V0.2 does not authorize polarimetry, measured-R mapmaking, JINC, maximum-likelihood mapmaking, mosaicking, or reprojection.

## 2 Linear operator, response, and absolute mode

For the selected frozen plan  $\Pi$ , transformed-sample input domain  $\mathcal{I}_\Pi$ , projection matrix  $G$ , diagonal coefficient matrix  $\Omega$ , neutral full-grid pixel vector  $\mathbf{v}$ , exact fixed-state upstream-to-PTC response  $H_{\text{fixed},\Theta}$ , and output-row selector  $J_{\text{out}}$ , the map operator, fixed-state response, and conditional covariance are:

$$D_Q = \text{diag}(G\Omega\mathbf{1}), \quad J_{\text{out}}\mathbf{v} = (v_p)_{p \in \mathcal{S}_{\text{out}}}, \quad D_{Q,\text{out}} = J_{\text{out}}D_QJ_{\text{out}}^\top, \quad (10)$$

$$A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1}J_{\text{out}}G\Omega, \quad \widehat{\mathbf{m}}_{\text{out}} = A_{\text{MAP},\Pi}\mathbf{z}, \quad (11)$$

$$\delta\mathbf{m}_{\text{out}} = A_{\text{MAP},\Pi}H_{\text{fixed},\Theta}\delta\mathbf{s}, \quad R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi}H_{\text{fixed},\Theta}, \quad C_{m,\text{out}} = A_{\text{MAP},\Pi}N_{\text{PTC}}A_{\text{MAP},\Pi}^\top. \quad (12)$$

The identity  $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q,\text{out}}^{-1}J_{\text{out}}G\Omega$  binds  $\mathcal{I}_\Pi$ , output-row domain  $\mathcal{S}_{\text{out}}$ , exact coefficient family/generation, `SCI-MAP:one_hot_containing_pixel01`, support policy, target WCS/grid, immutable parents, and lifecycle generation. The diagonal  $D_{Q,\text{out}}$  contains only finite positive normalizations on  $\mathcal{S}_{\text{out}}$ , so its ordinary inverse is defined. The matrix has no rows outside that set. Extending it to the full storage grid by writing zeros would be a representation convenience, not this scientific operator, response, or covariance.

Every valid MAP product discloses this complete realized  $A_{\text{MAP},\Pi}$  identity as its known MAP-local operator response. That disclosure is distinct from the upstream-conditioned response below and remains available when the latter is limited or unavailable.

Here  $N_{\text{PTC}}$  is a declared covariance of PTC-domain processed-sample error when such information is available, not the ensemble covariance of sky plus error. The state  $\Theta$  contains all selected or estimated calibration, alignment, membership, coefficients, response, geometry, identity, and effective-support facts. Equation (12) therefore makes a conditional perturbation statement on exactly the authorized rows; it makes no unnamed  $\mu_0$  or residual-bias claim. If any element of  $\Theta$  is estimated from the same data, an unconditional claim requires its joint uncertainty and cross-covariances.

The linear object  $H_{\text{fixed},\Theta}$  has an exact source domain, PTC-sample codomain, unit transformation, parent, membership, support, coefficient state, and response basis. It is not a name for a refitted procedure. When available,  $R_{\text{fixed},\Theta} = A_{\text{MAP},\Pi}H_{\text{fixed},\Theta}$  describes the response of that exact frozen state on the authorized output domain. A stored kernel image is the realized response to a named unit-bearing template  $\boldsymbol{\tau}$  only if its sample-domain parent is  $\mathbf{k} = H_{\text{fixed},\Theta}\boldsymbol{\tau}$  and the stored map is  $A_{\text{MAP},\Pi}\mathbf{k} = R_{\text{fixed},\Theta}\boldsymbol{\tau}$  with the same exact row identity as signal. Without that identity it is a tracer, not the measured response. If any signal member lacks the required exact parent, response is limited or unavailable on the unchanged signal membership; no hidden subset, zero, or identity array can stand in for it. A realized PTC-grid companion begins at MAP and receives  $A_{\text{MAP},\Pi}$  exactly once; it is never passed through upstream response again.

PTC full-procedure and PTC+MAP re-resolved responses are bounded finite-difference families, not one  $H$  matrix:

$$(\mathcal{D}_\Delta^z, \iota_0^z, \iota_+^z) = \text{authorize}_{\mathcal{C}_{\Delta^z}}(z_0, z_+), \quad \Delta\mathbf{z}_{\text{PTC-FP}}(\delta\mathbf{s}; \alpha) = \iota_+^z \mathcal{P}_{\text{PTC}}(\mathbf{s} + \delta\mathbf{s}; \alpha) - \iota_0^z \mathcal{P}_{\text{PTC}}(\mathbf{s}; \alpha), \quad (13)$$

$$\Delta\mathcal{S}_{\text{PTC-FP}} = \text{state}_{\text{PTC}}(\mathbf{s} + \delta\mathbf{s}; \alpha) \ominus \text{state}_{\text{PTC}}(\mathbf{s}; \alpha), \quad (14)$$

$$(\mathcal{D}_\Delta^m, \iota_0^m, \iota_+^m) = \text{authorize}_{\mathcal{C}_{\Delta^m}}(m_0, m_+), \quad \Delta\mathbf{m}_{\text{PTC-FP}} = \iota_+^m A_{\text{MAP},\Pi}\mathbf{z}_+ - \iota_0^m A_{\text{MAP},\Pi}\mathbf{z}_0, \quad \Pi \text{ fixed}, \quad (15)$$

$$\Delta\mathbf{m}_{\text{PTC+MAP-RR}} = \iota_+^m \mathcal{M}(\mathcal{P}_{\text{PTC}}(\mathbf{s} + \delta\mathbf{s}; \alpha); \Pi_+) - \iota_0^m \mathcal{M}(\mathcal{P}_{\text{PTC}}(\mathbf{s}; \alpha); \Pi_0), \quad (16)$$

A numerical full-procedure difference is defined only by the complete immutable `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. The record binds baseline and perturbed product identities; scientific quantity and unit; occurrence or MAP output-row identities; source and target frame/WCS; additive reference, gauge, and null-space state; baseline and perturbed support; coefficient and response basis; one exact common comparison domain; an exact map from each result into that domain; explicit treatment of rows present in only one result; perturbation amplitude, sign, side, location, source state, and normalization; finite-difference convention; state-transition record; and numerical availability with cause. It authorizes no implicit zero fill, union-domain sentinel, support intersection, coordinate-nearest or same-shape/row-position association, interpolation, reprojection, or gauge

alignment. Without the complete authorized record the numerical difference is unavailable, or the result is only a separately typed state transition/discontinuity.  $\Delta\mathcal{S}_{\text{PTC-FP}}$  never establishes subtractability by itself.

Every numerical procedure response shall bind that complete `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record defined in this section.

Here  $\alpha$  binds amplitude, sign, perturbation side, source state, location, normalization, domain, and every other family parameter.  $\Delta\mathcal{S}_{\text{PTC-FP}}$  separately records changed masks, groups, rank, loading subspace, classifications, coefficient state, stop cause, support, and other selected state. Equation (15) applies only when MAP membership, coefficient, projection, and support remain fixed. Equation (16) instead uses separately resolved  $\Pi_0$  and  $\Pi_+$ , which may produce different membership and output rows; it is the PTC+MAP re-resolved procedure response only after exact maps into the authorized common comparison domain are bound, not the fixed MAP operator applied to one matrix. The reserved whole-chain RTC-to-CAL-to-PTC-to-MAP response would rerun RTC, CAL, AST-dependent selection, PTC, MAP admission, coefficient/support resolution, and MAP from exact immutable parents under separate authority; it is unavailable unless separately established and is not claimed here. An unavailable upstream-conditioned response state does not invalidate the numerical map or prohibit later analysis. A later simulated response or response-corrected map is a new versioned product bound to the original MAP product and processing identity; it does not rewrite the original product's claim.

V0.2 fixes the signal-centering operator to  $L = I$ . SCI-MAP therefore does not remove a mean or a constant mode during coaddition. It preserves a constant admitted sample vector on the exact support-authorized output rows because numerator and denominator use identical coefficients. A stronger constant-sky statement additionally requires the forward operator to map that sky mode to the admitted sample constant. Neither result alone establishes point-source, integrated-flux, extended-source, edge, or absolute astrometric fidelity. The PTC nonrestored additive reference  $\lambda$ , fitted correlated removed component  $\hat{U}_{\text{corr}}$ , total removed component  $U_{\text{total},\Theta}$ , removed-subspace identity, fixed-state null space, full-procedure invariant/unidentifiable modes, and any separately authorized residual conditional bias are distinct facts. Any absolute mode already removed upstream remains absent and must be carried in response and null-mode provenance.

### 3 Input classes and falsifiable response behavior

The same response equation yields distinct, testable input classes without granting them equal scientific meaning:

**Constant.**

Fixed positive coefficients preserve a constant admitted sample vector even when coverage and variance vary.

**Delta or basis pixel.**

The output is a support-row-restricted column of  $R_{\text{fixed},\Theta}$ , including beam, projection, preprocessing, and boundary effects.

**Point-source template.**

The output is  $R_{\text{fixed},\Theta}\tau_b$  on exactly the authorized rows. Peak, integrated, and fit amplitudes are different functionals.

**Resolved, gradient, and Fourier structure.**

The output is  $R_{\text{fixed},\Theta}\tau$ ; constant preservation says nothing about these modes.

**Variable coverage.**

Fixed-coefficient normalization may preserve a constant while covariance and nonconstant response vary spatially.

**Finite boundary.**

Truncation can preserve a local constant yet change a compact-source peak or integral. No wrap or clamp is authorized.

The authorized base family is `SCI-MAP:one_hot_containing_pixel@1`. It uses half-open pixel cells, assigns unity to the unique finite in-grid containing pixel, excludes the outer upper boundary, never wraps or clamps, and has  $\sum_p G_{pi} = 1$  only on that in-grid domain. Fractional projection remains a future role; no kernel, conservation, boundary-loss, response, or covariance semantics are inferred for it.

Response fidelity is therefore a validation layer, not a consequence of code executing the algebra. Delta, point-template, extended, Fourier, variable- coverage, and edge injections are preregistered in Appendix D; they are not run or claimed here.

## 4 Conditional uncertainty and statistical labels

For independent selected sample errors with an exact selected coefficient family proven to be true inverse variance, the general formal relation is:

$$\text{Var}(\hat{m}_p \mid \Theta) = \frac{\sum_i G_{pi}^2 \gamma_i}{(\sum_i G_{pi} \gamma_i)^2}, \quad \text{Cov}(\hat{m}_p, \hat{m}_r \mid \Theta) = \frac{\sum_i G_{pi} G_{ri} \gamma_i}{Q_p Q_r}, \quad p, r \in \mathcal{S}_{\text{out}}, \quad (17)$$

$$\phi_p = \frac{Q_p^2}{\sum_i G_{pi}^2 \gamma_i}, \quad \phi_p = Q_p \iff \sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0, \quad p \in \mathcal{S}_{\text{out}}, \quad (18)$$

Equation (18) states why normalization is not precision by default. One-hot geometry satisfies the projection part of the equality, but cannot establish that  $\gamma$  is calibrated inverse variance or that sample errors are independent. The admitted uniform family has exact value one but explicitly supplies no precision or independence meaning.

The following objects must not be conflated:

1.  $Q$ , an ordinary normalization with its declared gridding unit;
2.  $\phi = 1/[C_{m,\text{out}}]_{pp}$ , formal conditional marginal precision under its named assumptions;
3.  $C_{m,\text{out}}$  or  $C_{m,\text{out}}^+$ , the full conditional covariance or precision on the identifiable stochastic subspace of the exact authorized row domain; and
4. NOI's exact `conditional_detector_sign_randomization_marginal_second_moment` and conditional scale, with only the claim authorized by that NOI method;
5. NOI standardized signal, whose sole current claim is MAP signal standardized by that conditional second-moment scale; and
6. any future calibrated-significance method, which requires separate exact authority and is unavailable here.

SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps  $Q$  normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

The bundle reports what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. Missing correlations are named, not set to zero. Neither normalization, hits, exposure, nor diagnostic weight is silently promoted to covariance. Absence of a complete covariance model does not invalidate the map or prohibit later scientific analysis. A later covariance estimate is a new versioned product bound to the original map and processing identity. The base map need not carry a complete covariance matrix or operator; it must disclose what actually persists and what remains lineage-resolvable or unavailable.

## 5 Distinct map facts, support, and MAP-local validity

The base/unfiltered MAP bundle separates facts because they answer different questions:

| Fact                                          | Meaning                                                                                                                                                                         |
|-----------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| AST-geometric incidence                       | Which coordinate-valid occurrences touch the unique containing pixel, independent of signal availability.                                                                       |
| MAP-route-candidate incidence                 | Which exact realized PTC occurrences also pass the occurrence-level MAP upstream-admission profile.                                                                             |
| Estimator-admitted contributions              | How many occurrences passed every ordinary coefficient and contribution predicate.                                                                                              |
| Admitted observation count                    | How many complete observation bundles contribute to a coadd pixel; not sample hits or attempted observations.                                                                   |
| Upstream-eligible original-footprint exposure | Unique original acquisition footprints ancestral to route candidates, placed by each original's own AST ALIGN-grid coordinate in the target WCS; not complete temporal support. |
| Retained original-footprint exposure          | The same original-footprint estimand restricted to ancestors of estimator-admitted membership; not precision or effective integration time.                                     |
| Normalization support                         | The numerical threshold predicate used to authorize ordinary division.                                                                                                          |
| Science-policy support                        | The separately stricter adopted policy predicate.                                                                                                                               |
| MAP-local base-product validity               | MAP's conjunction of admitted identity, support, finite signal, and no MAP-product failure; not downstream eligibility.                                                         |

No one fact is an alias for another. In particular, exposure is not precision, positive normalization is not final validity, and a finite payload is not evidence of scientific availability.

ALIGN owns  $e_{\text{acq}}$  and valid-original  $e_{\text{vo}}$ , in seconds on stable original occurrences. RTC, CAL, and PTC preserve those facts, exact lineage, and causes. MAP deduplicates by observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation. Under SCI-AST\_TO\_SCI-MAP\_ORIGINAL\_FOOTPRINT\_COORDINATE v0.1/r0.1, each positive- $e_{\text{vo}}$  original is placed exactly once using its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, with exact geometry/pointing realization, frame, validity, boundary, parent, and version. A descendant RTC-output coordinate is never substituted:

$$\mathcal{U}_p^{\text{up}} = \{u : e_u^{\text{vo}} > 0, G_p^u = 1, \exists i \in \mathcal{I}_{\text{route}} : u \rightsquigarrow i\}, \quad (19)$$

$$E_p^{\text{up,footprint}} = \sum_{u \in \mathcal{U}_p^{\text{up}}} e_u^{\text{vo}}, \quad (20)$$

$$\mathcal{U}_p^{\text{ret}} = \{u : e_u^{\text{vo}} > 0, G_p^u = 1, \exists i \in \mathcal{C} : u \rightsquigarrow i\}, \quad E_p^{\text{ret,footprint}} = \sum_{u \in \mathcal{U}_p^{\text{ret}}} e_u^{\text{vo}}, \quad (21)$$

$$\mathcal{U}_p^c = \bigcup_{o \in \mathcal{O}_p} \{(o, u) : u \in \mathcal{U}_{op}^{\text{ret}}\}, \quad E_p^c = \sum_{(o, u) \in \mathcal{U}_p^c} e_{ou}^{\text{vo}}. \quad (22)$$

The relation  $u \rightsquigarrow i$  is causal influence only. One original may influence several RTC/PTC descendants at different MAP pixels, but those edges do not relocate or duplicate its physical detector-seconds. Under the base one-hot family an original contributes exposure to at most one pixel, or none after boundary loss. Upstream-eligible exposure uses originals ancestral to the exact route-candidate population; retained exposure restricts that population to ancestors of estimator-admitted outputs. Coadd identity includes the observation, so union never merges distinct acquisitions. Overlapping filters, decimation, replacement, synthesis, donor reuse, cadence, hits,  $Q$ , and  $\gamma$  create no exposure. Missing exact original lineage or own-coordinate authority makes the affected exposure role unavailable rather than reconstructible from duration or count. Causal influence/support is published separately from representative RTC occurrence and physical original-footprint exposure. Neither exposure plane is complete temporal support, effective map integration time, precision, or normalized-map influence.

The adopted v0.2 threshold rule is operational policy authority, not a derived physical law. For the declared plane and policy population of finite positive  $Q_p$  values, sorted in ascending order, the admitted formula names  $c = \text{coverage\_cut}$ :

$$\mathcal{I}_{\Pi_{\text{eff}}} = \text{the declared support-policy population}, \quad (23)$$

$$\mathcal{P} = \{Q_p : \text{finite}(Q_p), Q_p > 0, p \in \mathcal{I}_{\Pi_{\text{eff}}}\}, \quad (24)$$

$$N = |\mathcal{P}|, \quad k = \left\lfloor \frac{\lfloor 0.75N \rfloor + N}{2} \right\rfloor, \quad (25)$$

$$Q_\star = \begin{cases} \mathcal{P}_{(k)}, & N > 0, \\ 0, & N = 0, \end{cases} \quad (25)$$

$$(c_{\text{req}}, c_{\text{eff}}, c_{\text{obs}}, c_{\text{app}}, c_{\text{real}}) = \text{requested, effective, observation-resolved, applied, and realized exact states and values}, \quad (26)$$

$$\mathcal{O}_{>1} = (I_{\text{request}}, I_{\text{authority}}, I_{\text{scope}}, c_{\text{req}}, c_{\text{eff}}, c_{\text{obs}}, c_{\text{app}}, c_{\text{real}}, \text{cause}, \text{provenance}), \quad (27)$$

$$o_{>1} = \text{resolved\_boolean}(\mathcal{O}_{>1}), \quad (27)$$

$$\text{admit}_{\Pi_{\text{eff}}}^{\text{pre}}(c_{\text{obs}}, o_{>1}) = \mathbf{1}[\text{finite}(c_{\text{obs}}), c_{\text{obs}} \geq 0, c_{\text{obs}} \leq 1 \vee o_{>1} = \text{true}], \quad (28)$$

$$c_{\text{app}} = c_{\text{obs}} \quad \text{only at the bound support-policy application event,} \quad (29)$$

$$c_{\text{real}} = c_{\text{app}} \quad \text{only when an immutable realized output records that application,} \quad (29)$$

$$T_{\text{norm}} = Q_\star c_{\text{app}} / 10, \quad T_{\text{sci}} = Q_\star c_{\text{app}}, \quad (30)$$

$$S_p^{\text{norm}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{norm}}], \quad (31)$$

$$S_p^{\text{sci}} = \mathbf{1}[\text{finite}(Q_p), Q_p > 0, Q_p \geq T_{\text{sci}}]. \quad (32)$$

Both predicates require the pixel's own explicit finite positive coefficient; the empty-population threshold of zero does not manufacture support. The operator row set  $\mathcal{S}_{\text{out}}$  is the exact intersection of the predicate row sets designated applicable by  $\Pi_{\text{eff}}$ ; it is not a full-grid mask with zero semantics.

The dimensionless  $c = \text{coverage\_cut}$  follows the five-stage MAP lifecycle. Pre-application validation checks its exact observation-resolved value; application binds that value to one attempt and a later realized product record exists only when actual immutable scientific output was produced. It is finite and nonnegative. Exact zero requests no relative cut; both thresholds become zero, but the row's own finite strictly-positive normalization remains mandatory. Values  $0 < c \leq 1$  are the ordinary recommended range. A value above one requires the independently authorized, same-value and same-scope Boolean plan fact **coverage\_cut\_above\_one\_override\_requested**; the numerical value does not authorize itself and may honestly leave science support empty. The 2026-09-08 owner directive selects for this r0.4 candidate the following empty-support disposition; final acceptance of these exact r0.4 bytes remains pending. If a valid applied support policy yields  $\mathcal{S}_{\text{out}} = \emptyset$ , or a valid applied coadd support policy yields  $\mathcal{S}_{\text{out}}^c = \emptyset$ , the application state is **applied\_no\_support\_authorized\_output\_rows**, the product and attempt outcome is **not\_produced**, and the exact cause is **no\_support\_authorized\_output\_rows**. The record retains request, effective, observation-resolved, and applied stages; exact parent and plan; coefficient, projection, support, role and profile identities;  $Q_\star$ , thresholds and support predicates; proof that the output-row set is empty; attempt identity, cause, and provenance. It has no realized MAP product or completion marker and publishes no dense zero map, zero-sky measurement, valid ordinary signal plane, no-row coadd member, or fabricated response/covariance plane. No immutable empty-row scientific-bundle alternative is authorized. This is a prospective rule for a future active route; this candidate asserts no current applied attempt. A negative, non-finite, missing, or unrepresentable value fails before support rows are constructed, application begins, or a required product is mutated. Each reached requested, effective, observation-resolved, applied, and realized state/value is retained; a successful realization has all five, while a stop before application, failure, or no-support nonproduction has only its reached prefix. The override record also binds request identity, authorized owner/profile, scope, exact value, cause, and provenance. No universal numerical default is supplied. Both threshold comparisons are inclusive, and neither support state implies exposure, precision, significance, response, completeness, or physical optimality.

## 6 Observation coaddition on one common grid

Coaddition consumes immutable complete logical observation bundles, not independent planes. Physical persistence follows the effective output plan; conforming in-memory and persisted bundles have the same scientific identity, admission, and arithmetic. A coadd-only entry may consume persisted conforming bundles without rerunning upstream processing. Every required publication still completes or fails, and this rule does not waive upstream PTC completion. The MAP-owned `SCI-MAP:observation_coadd_admission@2` is the proposed owner-governed aggregate-profile identity for this candidate. For the r0.4 generation, the proposed owner-governed MAP profile semantics are **candidate**; scientific-owner approval is **pending**; canonical Registry registration is **pending**; source binding is **pending**; SCI-VAL activation/evaluability is **unavailable**; and an object-specific decision realization is **unavailable**. In particular, `SCI-MAP:observation_coadd_admission@2` is the proposed owner-governed aggregate-profile identity for this candidate. After exact owner acceptance, canonical Registry/source binding, and activation, SCI-VAL may evaluate it for a requested object. The same candidate-generation separation applies to `SCI-MAP:map_upstream_admission@2`. Earlier exact profile approvals and evaluations retain their own immutable source generations; this candidate does not rebind or relabel them. Its future object-decision axes independently record request, applicability, eligibility, and decision-artifact realization; those axes are distinct from the MAP lifecycle and attempt outcome. The historical `SCI-MAP:observation_coadd_admission@1` record remains immutable and is not an alias for this role-aware successor. It atomically checks quantity and nominal-beam convention, PTC route/product role, response source/class, complete AST frame/WCS/grid, support policy, coefficient family, covariance disclosure, null/additive-reference state, required companions, lifecycle, and parentage before any coadd-owned state changes.

The six independently requestable product roles are `SCI-MAP:base_observation_map@1`, `SCI-MAP:response_bearing_observation_map@1`, `SCI-MAP:covariance_qualified_observation_map@1`, `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1`. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance.

| Role                             | Exact parent                                                                       | Required companions                                                                                                             | Missing/incompatible companion                                                                                                                                            |
|----------------------------------|------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Base observation                 | Exact realized PTC/AST inputs and complete MAP-local observation plan              | Complete MAP-local realized operator identity; honest upstream-response and covariance states                                   | Base signal may remain valid; state and cause are retained.                                                                                                               |
| Response-bearing observation     | Exact base-observation product named above                                         | Every named numerical response, with exact source domain, basis, class, unit, normalization, parent, WCS/grid, and row identity | This role fails or is unavailable before its required-product mutation; its base parent is unchanged.                                                                     |
| Covariance-qualified observation | Exact base-observation product named above                                         | Every within-observation covariance block named by the role on exact rows                                                       | Missing blocks are unknown and block this role only; its base parent is unchanged.                                                                                        |
| Base coadd                       | Ordered set of exact complete base-observation bundles and one resolved coadd plan | Every base signal companion and honest response and covariance identities/statuses                                              | Unavailable or basis-incompatible response and incomplete covariance are permitted with unchanged signal membership; coadd response becomes unavailable with exact cause. |

| Role                       | Exact parent                                                                    | Required companions                                                                                                                                                                                                                                                                                                                                 | Missing/incompatible companion                                                                                                                                                                                                                                                                       |
|----------------------------|---------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Response-bearing coadd     | Exact base-coadd parent named above and the identical ordered signal-member set | One exact response-family identity $R$ ; every member response $R_o^{(R)}$ ; compatible source domain, perturbation definition, basis, class, unit, normalization, reference/gauge/null state, parent, WCS/grid, and row identity; fixed membership, placement, $B_{\text{out}}$ , support and coefficients; and exact linear-composition authority | If the base parent already exists, this stronger role is unavailable for that parent; it never selects a subset. A separately requested admission may reject an observation before mutation only while constructing its own separately identified base/signal parent. No hidden subset or zero fill. |
| Covariance-qualified coadd | Exact base-coadd parent named above and the identical ordered signal-member set | Every named within- and cross-observation block on exact stacked rows                                                                                                                                                                                                                                                                               | One missing required block blocks this role for that parent; no member subset is formed. A separately requested admission may reject before constructing its own identified base/signal parent. Unknown is never zero or independence.                                                               |

The canonical target grid may be requested and made immutable before observation-map construction. The only v0.2 placement is exact centered integer embedding on the same grid. Observation and coadd row and column differences are nonnegative and even; their half-differences are the integer offsets, and the shifted reference pixels identify the same world coordinate. The lossless typed WCS is the authority. A later crop is a newly identified row-selected product with an immutable parent and lineage; it cannot relabel that parent as compatible. No fractional shift, post-hoc pad, interpolation, reprojection, mosaicking, wrapping, or implicit source recentering is allowed.

For the atomically admitted per-pixel set  $\mathcal{O}_p$ , every contributing observation row is already a member of that observation's support-authorized scientific domain. Let  $\mathcal{S}_{\text{out}}^c$  be the exact rows authorized by the effective coadd support policy. The MAP-owned family `SCI-MAP:uniform_observation_coadd_coefficient@1` assigns dimensionless  $u_{op} = 1$  to every admitted row. It is equal-observation weighting, not precision. The coadd is:

$$\mathcal{S}_{\text{out}}^c = \{p : \text{authorize}_{\Pi_{\text{eff}}^c}(p) = 1\}, \quad (33)$$

$$u_{op} = 1 \quad \text{for each admitted observation row,} \quad (34)$$

$$N_p^c = \sum_{o \in \mathcal{O}_p} u_{op} \hat{m}_{op}, \quad Q_p^c = \sum_{o \in \mathcal{O}_p} u_{op}, \quad (35)$$

$$\hat{s}_p = N_p^c / Q_p^c, \quad p \in \mathcal{S}_{\text{out}}^c, \quad (36)$$

$$J_{\text{out}}^c \mathbf{v}^c = (v_p^c)_{p \in \mathcal{S}_{\text{out}}^c}, \quad (37)$$

$$\hat{\mathbf{m}}_{\text{out}}^c = (\hat{s}_p)_{p \in \mathcal{S}_{\text{out}}^c} = B_{\text{out}} \hat{\mathbf{m}}_{\text{obs}}, \quad (38)$$

$$\hat{k}_p^{c,(R)} = \frac{\sum_{o \in \mathcal{O}_p} u_{op} \hat{k}_{op}^{(R)}}{Q_p^c}, \quad p \in \mathcal{S}_{\text{out}}^c. \quad (39)$$

$$C_{c,\text{out}} = B_{\text{out}} \Sigma^{\text{obs}} B_{\text{out}}^T \quad \text{only when every required block is available,} \quad (40)$$

$$T_{\text{obs}}^{(R)} = \text{ordered\_stack}_o \left( R_o^{(R)} \right),$$

$$R_{\text{coadd,out}}^{(R)} = B_{\text{out}} T_{\text{obs}}^{(R)},$$

only under the exact compatible fixed coadd state. (41)

Here  $\hat{\mathbf{m}}_{\text{obs}}$  is the stacked signal on exact observation-row identities;  $B_{\text{out}}$  is the centered-integer equal- observation averaging operator;  $T_{\text{obs}}^{(R)}$  stacks exact compatible member responses from one bound response family  $R$ ; and  $\Sigma^{\text{obs}}$  is the block covariance with every available within- and cross-observation block explicitly located. Equation (40) is a

complete numerical covariance claim only when every block required by the exact covariance-qualified role is available. Otherwise the state is explicitly partial, symbolic, summarized, lineage-resolvable, or unavailable. Unknown blocks are not zero or independence. The coadd operator  $B_{\text{out}}$  has rows only in  $\mathcal{S}_{\text{out}}^c$ . Coadd accumulators or dense storage slots outside that set do not become a zero-valued scientific coadd. On its exact domain the coadd is a convex weighted mean. One admitted observation reproduces itself; two admitted observations produce their equal-observation mean. Its normalization is not precision. With cross-observation covariance, the exact conditional covariance retains those blocks on the same selected rows. A correlated-GLS benchmark using  $(\Sigma_p^{\text{obs}})^{-1}\mathbf{1}$  may require negative combination coefficients, placing it outside this ordinary positive-coefficient contract.

For one exact response family  $R$ ,  $T_{\text{obs}}^{(R)}$  is the ordered stack of the exact compatible member responses  $R_o^{(R)}$ , and  $R_{\text{coadd,out}}^{(R)} = B_{\text{out}}T_{\text{obs}}^{(R)}$  only when every member has that same family identity; source domain, perturbation definition, basis, unit, normalization, reference/gauge/null state, WCS/grid and row maps are compatible; coadd membership, centered-integer placement,  $B_{\text{out}}$ , support and coefficient state are fixed for the query; and the family is mathematically composable through that fixed linear coadd operator. `SCI-MAP:response_bearing_coadd@1` binds the exact response-family identity it carries. Fixed-state, PTC full-procedure finite-difference, PTC+MAP re-resolved, coadd-re-resolved/full-procedure, and whole-chain responses are never one unqualified family. A finite difference is not multiplied through  $B_{\text{out}}$  as though it were a linear Jacobian without an exact composition theorem. A perturbation that may alter observation admission, coadd membership, support, placement, coefficients, role, or plan requires a separate coadd-full-procedure or coadd-re-resolved family and its own `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. Both that coadd-procedure response and the whole-chain response are unavailable until separately authorized.

The exact `SCI-MAP:base_coadd@1` permits unavailable or basis-incompatible response and partial or unavailable covariance with honest status. If any admitted member response is unavailable or basis-incompatible, it retains unchanged numerical signal membership and reports coadd response unavailable with exact cause; it forms no hidden subset or zero companion. The independently requested `SCI-MAP:response_bearing_coadd@1` and `SCI-MAP:covariance_qualified_coadd@1` each parent an exact base coadd with identical ordered membership and require only the response objects or covariance blocks named by that selected role. A required incompatibility blocks the stronger role for an existing parent and never subsets or rewrites it. A separately selected stronger-role admission may reject an observation atomically only while constructing its own separately identified base/signal parent and before any of that state mutates. Retained exposure and admitted observation count follow the same atomic bundle admission, while exposure retains its unique-original own-coordinate placement. A rejected observation changes no coadd state anywhere, even if a subset of its planes would otherwise appear compatible.

## 7 Identity, WCS, units, and lifecycle

A map cannot be reconstructed from its collection position. Paired PTC  $x/r$  components remain upstream detector coordinates and are carried only when a MAP-facing fact requires their identity; MAP does not recast  $r$  as a response. The minimum identity includes the nonpolarimetric total-intensity-equivalent product role; external observation or coadd ID; array/group and occurrence domains; estimator, projection, PTC and coadd coefficient families; MAP-local operator and upstream-conditioned response/covariance states; signal unit and exact nominal-beam parent; complete AST frame/WCS; preconstruction target-grid identity where applicable; exposure convention; logical-bundle and persistence state; MAP profiles; exact PTC/AST generations and product identity. A STOKES field is only a locator unless separate quantity authority exists. The policy must be scientifically identifiable, durable, and comparable, but this contract prescribes no JSON, hash, or sidecar serialization. Array ID, name, index, network, map group, map index, detector occurrence, and container position remain distinct.

Science maps bind the complete frame identity supplied by `SCI-AST:rtc_output_grid_coordinates@1`, including the actual reference system, equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued axes. MAP never equates J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name/default. Pointing and OOF may reuse the exact ordinary MAP gridding operator when their inputs satisfy this contract. Their target WCS/frame, source model, fitting, response interpretation, outputs, defaults, profiles, and qualification remain mode- owned; no unsupplied mode profile becomes available. FITS/WCS coordinates are one-based and in-memory row/column indices are zero-based.

Full-precision typed or sidecar WCS is the lossless admission and provenance authority. For an immutable finite MAP grid of  $n_x$  by  $n_y$  cells, the 0.1-arcsec representation-test population is the finite union of (a) every scientifically addressable integer pixel center  $(j+1, k+1)$  for  $0 \leq j < n_x$ ,  $0 \leq k < n_y$ , and (b) every outer-footprint cell vertex:  $(h + \frac{1}{2}, \frac{1}{2})$ ,  $(h + \frac{1}{2}, n_y + \frac{1}{2})$  for  $0 \leq h \leq n_x$ , together with  $(\frac{1}{2}, l + \frac{1}{2})$  and  $(n_x + \frac{1}{2}, l + \frac{1}{2})$  for  $0 \leq l \leq n_y$ . The second set includes all four outer corners and every required outer-boundary vertex. The complete population is preregistered from the full-precision typed WCS, grid shape, and row domain before the physical FITS WCS is generated. Both representations shall bind the same WCS-generation identity, celestial frame, axis order, signs, handedness, orientation, shape, and centered-integer reference-pixel relation. At every population member, each WCS is evaluated at the exactly corresponding center (zero-based memory index  $(j, k)$ , one-based FITS pixel  $(j+1, k+1)$ ). Separation is the great-circle angle  $\text{atan2}(\|\mathbf{u} \times \mathbf{v}\|, \mathbf{u} \cdot \mathbf{v})$  between unit direction vectors in that exact declared frame, evaluated with IEEE-754 binary64 round-to-nearest, ties-to-even operations and reported in arcseconds. The maximum over the complete finite population shall be at most 0.1 arcsec. If either WCS yields an invalid or non-finite direction at any preregistered member, the representation fails; the point is not dropped. A center is absent only when its scientific row was already absent before population generation; every listed outer-boundary vertex remains required. No smaller sampled subset or continuum/global-sky claim is authorized. This is a representation tolerance, not authority for a different grid.

The complete ordered MAP lifecycle, its independent attempt outcome, and its record rules appear once in the following lifecycle table.

| MAP stage                   | Exact meaning                                                                                                                                                                | Record rule                                                                                                |
|-----------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------|
| <b>requested</b>            | Immutable accepted intent.                                                                                                                                                   | Exists only for an accepted request; it is never reconstructed from a later stage.                         |
| <b>effective</b>            | Supported executable interpretation of that request.                                                                                                                         | Binds supported/disabled/unavailable interpretation without mutating intent.                               |
| <b>observation_resolved</b> | Every observation-specific parent, coefficient, profile, geometry, mapping, parameter, support rule, output role, and persistence decision is concrete or typed unavailable. | Supplies the exact value/state for pre-application validation.                                             |
| <b>applied</b>              | Exact resolved operation and parameter state was invoked or committed for one execution attempt.                                                                             | Created only at the application event; retained with exact terminal failure when output realization fails. |
| <b>realized</b>             | Actual immutable accumulators, scientific output rows, product values, role-qualified companion statuses, product identity, and provenance were produced.                    | Created only for an actual MAP product; a failure- evidence record is not product realization.             |

| MAP stage        | Exact meaning                                                                                                        | Record rule                                                                                                                                                                                                 |
|------------------|----------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Attempt outcome  | Exactly <b>succeeded</b> , <b>failed</b> , or <b>not_produced</b> .                                                  | Recorded independently of the reached lifecycle prefix and product realization.                                                                                                                             |
| Termination rule | Stop before application; post-application terminal failure; or applied valid policy with no support-authorized rows. | The first has no applied or realized record. The latter two retain the applied-attempt record and exact cause but no realized MAP product; the no-support case is <b>not_produced</b> , not <b>failed</b> . |

The MAP plan, coefficient selection/application, projection, support policy, output/coadd plan, response/covariance request, product role, and **coverage\_cut** use this same ordering. Each parameter stage binds its exact value and provenance; a completed realization contains all five stages, while a nonproducing attempt contains only stages actually reached.

The MAP lifecycle does not replace the independently named PTC/VAL handoff axes **requested/not\_requested**, **applicable/inapplicable/applicability\_unknown**, **eligible/ineligible/decision\_unavailable**, and **realized/incomplete/failed/not\_produced**. It also does not replace the ECS requirement-result vocabulary or evidence- artifact realization state. In particular, a MAP parameter's resolved or applied value may be checked before application or bound after application; its **realized** value is never a prerequisite for a pre-application check.

Resolution, application, and realization are distinct. Pre-application validation examines the exact **observation\_resolved** parameter and profile state. Application binds that resolved state to one attempt. A later realized product record exists only when actual immutable MAP output was produced; attempt failure evidence does not realize the product and cannot be used retroactively as the prerequisite that authorized the attempt. The independent outcome axis follows [the separate MAP attempt-outcome rule](#).

## 8 Failure, downstream boundary, and immutable parentage

Failure semantics protect the scientific bundle from partial truth. An invalid candidate can be excluded at contribution scope; an unsupported pixel is MAP-local invalid; an incompatible observation is rejected atomically only at the scope of the exact selected role and its required facts; an unrepresentable finite aggregate, coordinate/index conversion, missing role-required companion, or failed required publication propagates at its declared product scope before live state mutation. A pre-application stop has no applied or realized product record; a terminal failure after application retains the applied attempt and immutable failure evidence without a fabricated realized product. A valid applied no-support outcome instead retains the exact attempt, proof, outcome **not\_produced**, and cause **no\_support\_authorized\_output\_rows** without asserting failure. An unavailable response or covariance inside a permitted realized base bundle is a companion-status fact, not attempt failure. A completion marker cannot override a missing required output.

No downstream package receives a bare signal plane as a complete map. The boundary includes the estimator and normalization identity; response state; conditional covariance/formal-weight state and assumptions; distinct map facts; additive-bias and null-mode state; units; WCS; grouping; lifecycle; base/unfiltered MAP parentage; dependency versions; and failures. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps  $Q$  normalization, formal conditional precision, complete conditional covariance, the NOI **conditional\_detector\_sign\_randomization\_marginal\_second\_moment**, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims. NOI may propagate a fixed operator only under its exact method. FLT owns filtered support, response, covariance, and validity. SRC, MODE, and BEAM own inference and fit uncertainty. FRUIT owns recurrence and iteration state.

Every later analysis is responsible for deciding which facts its own claim requires. An unavailable response prevents the original MAP bundle from making an unsupported response-dependent claim, but it does not prohibit analysis or invalidate other map claims. A significance analysis cannot relabel merely formal standardization; an absolute-level claim cannot ignore an unobservable offset. This contract defines no closed-world registry of all future uses. A downstream product may have local validity or eligibility, but it cannot rewrite MAP-local base-product validity or sever the immutable base/unfiltered MAP-parent identity.

When an admitted noise realization is described as using the same operator, membership, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and coadd admission are fixed or identically prescribed. Recomputing them from randomized data creates a different estimator, regardless of matching array shapes or filenames.

## 9 Uncertainty, validity, validation, and claim layers

The contract deliberately separates four reasoning layers:

1. **Uncertainty layer:** conditional covariance and response are propagated through the fixed operators; omitted state uncertainty remains explicit.
2. **Validity layer:** contribution eligibility, numerical support, science-policy support, and MAP-local base-product validity determine the exact rows on which the estimator is scientifically available; unsupported rows are absent, not zero-filled observations.
3. **Validation layer:** preregistered analytic fixtures, matrix comparisons, injections, covariance controls, representation round trips, and observational references test distinct claims.
4. **Readiness layer:** implementation evidence, operational reliability, data completeness, and release authority are assessed elsewhere.

| Claim layer                      | Status here           | Evidence needed before an achieved claim                                                                              |
|----------------------------------|-----------------------|-----------------------------------------------------------------------------------------------------------------------|
| Algebraic scientific contract    | Proposed author draft | Scientific-owner review and contract freeze; analytic predictions are specified, not executed.                        |
| Engineering conformance          | Not assessed          | Requirement-by-requirement inspection and tests against one exact candidate identity.                                 |
| Representation/response fidelity | Not assessed          | WCS round trips and delta, template, extended, variable-coverage, and edge injections with preregistered bounds.      |
| Observational performance        | Not assessed          | Independent astronomical amplitude, shape, position, covariance, and repeatability references where available.        |
| Production readiness             | Not assessed          | Complete required artifacts, failure/log audit, operational gates, owner release decision, and no disqualifying gaps. |

Floating reductions require a preregistered bound derived from operation count and conditioning. For a sum of  $n$  terms, a reference policy may use  $\gamma_n = n\epsilon/(1 - n\epsilon)$  and a fixed multiple such as the admitted core's  $8\gamma_n \sum_i |a_i|$ , frozen before observing results. Integer counts, identities, masks, support, offsets, and product cardinality remain exact. Tolerances are not relaxed after failure.

The contract's 25 predictions are listed formally in Appendix D. No validation run, implementation inspection, observational reduction, or production decision was performed during this scientific authorship stage.

## 10 Owner decisions, external hard gates, and route disposition

All nine stable MAP decisions are decided for successor drafting. OD-001–007 and OD-009 use the approved 2026-08-28 dispositions; OD-008 retains the exact one-hot half-open projection. PTC-OD-010 supplies the versioned coefficient- Registry architecture, and the owner-approved uniform authority supplies one exact family, handoff profile, and separate MAP permission. These scientific decisions do not by themselves create an actual selection, publication, profile evaluation, application route, Registry activation, validation, or freeze.

| Route level                           | Contract status                                  | Exact remaining gate                                                                                                                                                                                                                                                                                                                                                                          |
|---------------------------------------|--------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Conditional estimator authority       | Defined                                          | Positive-coefficient observation and equal-observation centered-integer coadd equations are conditional scientific authority.                                                                                                                                                                                                                                                                 |
| Dependent numerical observation route | <b>UNAVAILABLE in this candidate</b>             | Requires an explicitly selected admitted family, complete PTC transformed-product publication and exact handoff, admitted finite-nonnegative <b>coverage_cut</b> , every independent gate, and owner-accepted/canonically registered/source-bound and active successor profiles. The exact r0.4 MAP and VAL records are candidate proposals; no object decision or realized route is claimed. |
| Response-bearing science map          | Conditional or unavailable                       | The MAP-local realized operator is disclosed; stronger claims require one exact sufficient fixed-state, PTC full-procedure, or PTC+MAP re-resolved family. A response-bearing coadd binds one exact compatible family and composition theorem. Coadd-full-procedure/re-resolved and whole-chain responses remain separately unavailable.                                                      |
| Uncertainty-qualified map             | Conditional or unavailable                       | Requires the exact covariance representation and every term/block needed by the named claim; unknown is not zero.                                                                                                                                                                                                                                                                             |
| Compatible-grid coadd                 | Defined conditionally; current route unavailable | Requires exact admitted observation bundles plus owner-accepted, canonically registered/source-bound, active <b>observation_coadd_admission@2</b> and a realized object decision; response/covariance qualification remains role-specific. No source-closed frozen numerical route is claimed.                                                                                                |
| Future reprojection/mosaic            | Not authorized                                   | A separate scientific contract remains required; preconstruction target-grid request and exact centered-integer embedding do not authorize resampling.                                                                                                                                                                                                                                        |

| External hard gate         | Contract disposition                                    | Route effect                                                                                                                                                                           |
|----------------------------|---------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| PTC MAP-facing coefficient | Uniform family admitted; actual handoff input-dependent | Registry presence alone realizes nothing. Missing selection, complete PTC publication, exact association, permission, or four-axis handoff blocks the affected route with no fallback. |

|                                                                     |                                                                                                      |                                                                                                                                                                                                                                                                                                                                       |
|---------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Admitted numerical<br><code>coverage_cut</code> state               | Scientific domain closed;<br>actual state<br>input-dependent                                         | Bad or missing resolved state fails before<br>support construction or mutation; > 1<br>additionally requires the independently<br>authorized same-value/scope <code>coverage_cut_</code><br><code>above_one_override_requested</code> fact.                                                                                           |
| PTC-to-MAP uniform<br>boundary<br>MAP upstream-admission<br>profile | Approved exact uniform<br>successor<br>Exact r0.4 candidate<br>generation; activation<br>unavailable | The binding names the exact boundary ID and<br>digest/version; a mismatch blocks the route.<br>Scientific semantics are candidate; owner<br>acceptance and canonical Registry/source<br>binding are pending; evaluability and object<br>decision are unavailable. Earlier<br>approvals/evaluations retain their exact<br>generations. |
| Exact AST same- <i>n</i> coordinate<br>parent                       | Contract role closed;<br>realization<br>input-dependent                                              | Missing/mismatched parent excludes the<br>occurrence.                                                                                                                                                                                                                                                                                 |
| Original-footprint exposure<br>lineage and AST ALIGN-grid<br>parent | Exact boundary closed;<br>realization<br>input-dependent                                             | Missing stable-original, target-WCS, validity,<br>or boundary facts make required exposure<br>unavailable and block a product that<br>designates it required.                                                                                                                                                                         |
| Response/covariance state                                           | Claim-dependent                                                                                      | Does not by itself invalidate the base signal;<br>blocks only a selected role that names the<br>unavailable exact object or block as required.                                                                                                                                                                                        |

The contract is coherent at this conditional boundary. A scientific route is defined only when every actual structural, selection, publication, handoff, admission, numerical, coordinate, support, and lifecycle gate passes. The candidate establishes no implementation conformity, response fidelity, validation, observational performance, freeze, readiness, Registry activation, application availability, or production authorization.

## A Owner-decision register

This compact register is generated from `SCIENTIFIC_OWNER_DECISION_LEDGER.md`. The mechanical contract checker requires the nine exact IDs and decided dispositions to match in all views. Rendering this register does not freeze or activate the candidate.

| Decision/status                  | Approved disposition                                                                                                                                                                                                  | Binding consequence                                                                                                                                     |
|----------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| SCI-MAP-0D-001<br><b>DECIDED</b> | Relative support is an operational policy, with no exposure, precision, significance, response, completeness, or physical-optimality meaning.                                                                         | Preserve the exact population, order statistic, thresholds, and policy identity.                                                                        |
| SCI-MAP-0D-002<br><b>DECIDED</b> | Independent experimentation is unrestricted; official identity and claims follow mathematical equivalence or versioned change control with appropriate evidence.                                                      | No permission process for research and no retroactive relabeling.                                                                                       |
| SCI-MAP-0D-003<br><b>DECIDED</b> | Every valid map discloses its known MAP-local realized operator. Upstream- conditioned and whole-chain responses remain separately typed and honest.                                                                  | Unsupported response-dependent claims remain unavailable; later estimates are new versioned products bound to the original.                             |
| SCI-MAP-0D-004<br><b>DECIDED</b> | Report actual covariance information, domain, assumptions, omissions, limitations, and lineage; a complete covariance object is not mandatory.                                                                        | Unknown is never zero or independence; later estimates have new identity.                                                                               |
| SCI-MAP-0D-005<br><b>DECIDED</b> | A complete logical observation bundle precedes coaddition; physical persistence follows the effective plan.                                                                                                           | In-memory and persisted/coadd-only routes use identical admission, arithmetic, identity, and provenance; required publication still completes or fails. |
| SCI-MAP-0D-006<br><b>DECIDED</b> | Pointing and OOF may reuse the exact ordinary MAP operator with conforming inputs.                                                                                                                                    | Mode WCS/frame, model/fitting, interpretation, outputs, defaults, profiles, and qualification remain with their owners.                                 |
| SCI-MAP-0D-007<br><b>DECIDED</b> | <code>coverage_cut</code> is finite and nonnegative: zero gives no relative cut; $(0, 1]$ is ordinary; $> 1$ requires an explicit expert request; bad or missing values fail before support construction or mutation. | Retain requested, effective, observation-resolved, applied, and realized states/values; no universal default.                                           |
| SCI-MAP-0D-008<br><b>DECIDED</b> | The one-hot lower-inclusive, upper-exclusive containing-pixel family remains the only ordinary projection.                                                                                                            | Boundary loss contributes nowhere; fractional projection remains deferred.                                                                              |
| SCI-MAP-0D-009<br><b>DECIDED</b> | A canonical target grid may be requested before construction; compatible maps use exact centered-integer embedding.                                                                                                   | A later crop is a new row-selected product; fractional resampling, reprojection, and mosaicking remain outside this contract.                           |

## B Requirement and crosswalk summary

The detailed 52-row requirement crosswalk and 25-row prediction crosswalk remain in `CROSSWALK.md`. This compact index makes the rationale's traceability visible without duplicating the normative clause text or the detailed row table. Appendix C renders every canonical requirement, and Appendix D renders every prediction, directly from the shared authority.

| Authority range | Scientific surface                                                                                           | Principal rationale / remaining gate                               |
|-----------------|--------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------|
| REQ-001         | Version and revision identity                                                                                | Cover; owner freeze                                                |
| REQ-002–010     | Capability, input, identity, projection, coefficients, lifecycle, and admission                              | Sections 1, 7, 8; upstream producer facts                          |
| REQ-011–018     | Ordinary accumulators, exact support-row publication, response, and absolute mode                            | Sections 1–3; actual response input                                |
| REQ-019–024     | Row-selected covariance, formal weight, nuisance state, and statistical labels                               | Sections 2, 4; claim-specific covariance input                     |
| REQ-025–035     | Separate incidence/contribution/exposure/support facts, threshold policy, exact rows, and MAP-local validity | Section 5; actual cut and support state                            |
| REQ-036–042     | Complete bundles, atomic centered-integer coadd, row-selected coadd                                          | Sections 6, 8; actual complete bundle and role-specific companions |
| REQ-043–048     | response/covariance, and GLS boundary                                                                        |                                                                    |
| REQ-043–048     | Product identity, WCS, provenance, failure, and publication                                                  | Sections 7–8; actual plan and mode/profile authority               |
| REQ-049–052     | Immutable base MAP parent, consumer boundary, fixed realization operator, and claim separation               | Sections 8–9; consumer-specific claim gates                        |
| PRED-001–025    | Analytic consequences and conformance/representation challenges                                              | Sections 3, 9; Appendix D; not executed                            |
| OD-001–009      | Stable scientific-owner decisions                                                                            | Section 10; Appendix A; all nine DECIDED for successor drafting    |
| SCI-MAP-CI-001  | Resolved dimensionless classification and bounded amendment                                                  | Section 5; equations, REQ-031/032, and PRED-012 amended in place   |

## C Canonical requirements

This appendix is rendered directly from the shared canonical LaTeX authority. Requirement identifiers are stable within SCI-MAP v0.2. Editing prose in this document does not silently change them.

**SCI-MAP-REQ-001 – Versioned authority.** The scientific contract version shall be v0.2. The rendered document revision shall be recorded separately as r0.4; changing layout or exposition alone shall not be represented as a new scientific contract version.

**SCI-MAP-REQ-002 – Capability and realized PTC input.** The ordinary input shall be the realized SCI-PTC v0.1/r0.5 transformed calibrated- $x$  product  $z_i = Z_i^{\text{PTC}}$ , a nonpolarimetric total-intensity-equivalent detector-time quantity in the inherited top-of-atmosphere, point-source-equivalent mJy/beam convention with its exact originating fixed-nominal-beam identity. Neither that unit nor a STOKES token establishes Stokes I. MAP has no CAL-to-MAP fallback, no inferred no-op PTC product, and no product when the required positive-rank PTC product is absent or PTC is disabled.

**SCI-MAP-REQ-003 – Exact occurrence identity.** Every candidate occurrence  $i$  shall bind observation, detector occurrence and UID, stable RTC output sample  $n$ , exact PTC product/application generation, segment, and array/network/group as applicable. Sample time is an associated attribute. Input-column, row, and container positions are locators only and shall not establish identity or equality.

**SCI-MAP-REQ-004 – Producer facts and MAP admission.** SCI-MAP shall consume PTC-owned transformed-signal availability, output-retention disposition, coefficient family/value/typed availability, and coefficient/QC proposition without reinterpretation. The PTC coefficient/QC proposition passes only when the coefficient is requested, applicable, eligible, and realized under the exact PTC-owned coefficient/QC profile for the declared MAP-facing family. It does not grant MAP use. SCI-MAP shall separately apply `SCI-MAP:map_upstream_admission@2` only after its exact owner acceptance, canonical Registry/source binding, and activation. [the exact candidate profile-status rule](#) applies to this generation. VAL Core may then evaluate that MAP-authored policy for an exact requested object but does not author it. The immutable r0.5 profile `SCI-MAP:map_upstream_admission@1` is historical, is not a compatibility alias, and shall not replace `SCI-MAP:map_upstream_admission@2`. Every fact, decision, cause, scope, and immutable reason remains separately typed.

**SCI-MAP-REQ-005 – Coordinate and projection authority.** MAP shall consume `SCI-AST:rtc_output_grid_coordinates@1` for the same stable RTC sample  $n$  and complete parent chain as  $z_i$ ; it shall never join by time, row, cardinality, or coordinate equality. MAP owns  $G_{pi}$  through `SCI-MAP:one_hot_containing_pixel@1`: the unique finite in-grid pixel whose cell contains the coordinate under half-open bounds contributes  $G_{pi} = 1$ , every other pixel has  $G_{pi} = 0$ , and an outer-upper-boundary or out-of-grid coordinate contributes nowhere. Fractional projection is deferred. AST may materialize the relation only when the artifact binds the AST parent and exact MAP plan.

**SCI-MAP-REQ-006 – Ordinary coefficient status.** The ordinary route requires one explicitly selected PTC-owned MAP-facing analysis/gridding coefficient family satisfying SCI-PTC-REQ-052–055 and `SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4`. The historically owner-approved and source-bound Registry `SCI-PTC:analysis_gridding_coefficients@draft-0.1` contains `SCI-PTC:uniform_constant@draft-0.1`, whose exact supported value is dimensionless  $\gamma_i = \omega_i = 1$ , and grants MAP permission through the historically owner-approved `SCI-PTC:uniform_coefficient_handoff@draft-0.1`. Those exact approvals and literal identifiers remain immutable; they neither activate this r0.4 candidate generation nor realize an object decision. Registration and permission do not select a family or realize a handoff. MAP shall use this family only after explicit selection; one immutable completed SCI-PTC v0.1/r0.5 transformed-product publication bound to the exact application, CAL parent, observation, and selection lineage and typed as a complete segment, complete scan, or bounded chunk wholly within one named scan or segment and one observation; exact target/payload/occurrence association; and a requested, applicable, eligible, realized profile evaluation under an exact active compatible binding generation. MAP does not reproduce, select, or infer the PTC-owned formula. An absent selection without an authorized mode default, unregistered family, inactive dependent binding, incomplete PTC publication, missing or mismatched payload, unavailable coefficient/QC decision, or incompatible generation makes the affected route unavailable; there is no unity or alternate-family fallback. Structural availability remains distinct from finiteness. MAP alone classifies an authorized payload as finite positive, exact zero, negative finite, non-finite, or unrepresentable and applies only a matching finite strictly-positive value. The uniform family supplies no precision, sensitivity, loading, scatter, response, covariance, exposure, validity, optimality, or default meaning.

**SCI-MAP-REQ-007 – Exposure lineage and accounting.** ALIGN-owned  $e_{acq}$  and valid-original  $e_{vo}$ , in seconds, shall remain immutable on stable original occurrences through RTC/CAL/PTC. The products are normatively `upstream_eligible_original_footprint_exposure` and `retained_original_footprint_exposure`. The shorter prose labels “upstream-eligible exposure” and “retained exposure” are explicit aliases for those products and no others. MAP shall deduplicate by exact observation, detector occurrence/UID, stable original/native occurrence, stable ALIGN slot, and ALIGN mapping generation; place each positive- $e_{vo}$  original exactly once through `SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1`, its layered AST ALIGN-grid direction/tangent/continuous-pixel parent in the exact target MAP WCS, and `SCI-MAP:one_hot_containing_pixel@1`; and never use a descendant RTC-output coordinate. Geometry/pointing realization, frame, coordinate validity, boundary state, parent, and version are required. Upstream-eligible and retained exposure differ only by ancestor-population restriction. Causal influence, representative RTC occurrence, and original-footprint exposure remain separate. These exposure products are not complete temporal support, effective map integration time, precision, or normalized-map influence. Missing original-coordinate authority makes the affected exposure role unavailable. Invalid-original has  $e_{vo} = 0$ ; synthesis, replacement, donor reuse, overlapping filters, and decimation create no exposure. Boundary loss contributes nowhere. Coadd exposure unions atomically admitted observation-scoped original identities.

**SCI-MAP-REQ-008 – Response information state.** Every valid `SCI-MAP:base_observation_map@1` product shall disclose its known MAP-local realized operator response  $A_{MAP,\Pi}$ , including exact input/output domains, membership, projection, coefficient, support-row, WCS, parent, lifecycle, and limitations. MAP shall separately type upstream-conditioned fixed-state linear response, PTC full-procedure finite differences, and PTC+MAP re-resolved procedure response. An available fixed-state object shall also bind source domain, PTC-sample codomain, units, parents, response basis, and upstream limitations. Every numerical full-procedure  $\Delta z_{PTC-FP}$ ,  $\Delta m_{PTC-FP}$ , or  $\Delta m_{PTC+MAP-RR}$  shall bind the complete `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. A numerical full-procedure difference is defined only by the complete immutable `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. The record binds baseline and perturbed product identities; scientific quantity and unit; occurrence or MAP output-row identities; source and target frame/WCS; additive reference, gauge, and null- space state; baseline and perturbed support; coefficient and response basis; one exact common comparison domain; an exact map from each result into that domain; explicit treatment of rows present in only one result; perturbation amplitude, sign, side, location, source state, and normalization; finite- difference convention; state-transition record; and numerical availability with cause. It authorizes no implicit zero fill, union-domain sentinel, support intersection, coordinate-nearest or same-shape/row-position association, interpolation, reprojection, or gauge alignment. Without the complete authorized record the numerical difference is unavailable, or the result is only a separately typed state transition/discontinuity.  $\Delta S_{PTC-FP}$  never establishes subtractability by itself. The PTC+MAP family reruns PTC and separately resolves  $\Pi_0$  and  $\Pi_+$ , whose membership and  $S_{out}$  may differ; subtraction is unavailable without exact maps into one authorized common comparison domain. A coadd-re-resolved/full-procedure response and a whole-chain RTC-to-CAL-to-PTC-to-MAP response each require separately authorized complete procedures and remain unavailable unless so established. Absent upstream response shall be reported as limited or unavailable without concealing the known MAP-local operator, invalidating an otherwise valid base map, fabricating a procedure/whole-chain identity, or prohibiting later analysis. A later response estimate or corrected map is a new immutable versioned product bound to the original MAP product and processing identity.

**SCI-MAP-REQ-009 – Immutable plan, lifecycle, and attempt outcome.** [the exact MAP lifecycle rule](#) and [the separate MAP attempt-outcome rule](#) apply. The same ordered lifecycle shall bind the MAP plan, coefficient selection and application, projection, support policy, output plan, coadd plan, response and covariance requests, every independently requested product role, and `coverage_cut`. [the exact lifecycle and independent-axis separation rule](#) applies. For a valid applied policy with no support-authorized output rows, [the exact empty-support disposition](#) applies. Disabled expert values remain inactive; one observation’s resolved state shall not leak into another.

**SCI-MAP-REQ-010 – Factored contribution predicate.** For each pixel, membership in  $C_p$  shall be the conjunction of ten exact Boolean pass predicates derived from separately retained typed records: transformed-signal availability; PTC output retention; coefficient availability; PTC coefficient/QC proposition pass $_{\gamma,QC}$ ; MAP upstream admission; AST coordinate validity; finite  $z_i$ ; finite strictly positive  $\gamma_i$ ; admitted projection/boundary relation; and MAP-local contribution plus required-companion predicates. Every VAL-evaluated gate retains named request, applicability, eligibility, and realization fields; only request=`requested`, applicability=`applicable`, eligibility=`eligible`, and realization=`realized` passes. Every other field combination is nonpassing with its exact cause, failure scope, and immutable reason retained. A structural source- or profile-binding failure is recorded only when that structural condition actually occurs; an ordinary nonpassing field value is not thereby a structural failure.

Evaluation follows the ordered DAG: A structural identity/parent, generation, source/profile, producer availability, and decision-realization checks; B retrieval only of structurally admitted payloads and MAP classification of  $z_i$  and  $\gamma_i$ ; C AST validity, one-hot boundary/placement, and MAP-local required companions; D membership construction; E accumulation. All gates pass before payload arithmetic. Reading an authorized payload to classify it is not arithmetic, and excluded non-finite values never contaminate accumulation. T/F/U/C or four-field records shall never be conjoined directly; no numeric-zero mask represents exclusion.

**SCI-MAP-REQ-011 – Ordinary accumulators.** On the fixed contribution set, SCI-MAP shall accumulate  $N_p = \sum_i a_{pi} z_i$ ,  $Q_p = \sum_i a_{pi}$ , and, when available for the identical membership,  $K_p = \sum_i a_{pi} k_i$ , where  $a_{pi} = G_{pi} \gamma_i > 0$ .

**SCI-MAP-REQ-012 – Normalized publication.** The scientific normalized output vector shall contain exactly the rows  $p \in \mathcal{S}_{\text{out}}$  authorized by the applicable effective support policy. On those rows finite  $Q_p > 0$  gives  $\hat{m}_p = N_p/Q_p$ , and an available response companion gives  $\hat{k}_p = K_p/Q_p$ . A finite exact-zero signal numerator  $N_p = 0$  is admissible when all other gates and support facts pass and gives the valid normalized scientific value  $\hat{m}_p = 0$ . No unsupported row, fabricated zero-filled substitute, or sentinel is a normalized scientific output or measured zero. If  $\mathcal{S}_{\text{out}} = \emptyset$  after a valid applied support policy, [the exact empty-support disposition](#) applies.

**SCI-MAP-REQ-013 – Accumulator interpretation.** The numerator is a pre-normalized accumulator and  $Q_p$  is a normalization fact by default. Neither shall be labeled a sky estimator, inverse variance, confidence, coverage, or significance without the separately required conditions.

**SCI-MAP-REQ-014 – Constant preservation.** With fixed admitted membership and positive coefficients, the ordinary estimator shall preserve a constant admitted sample input on every row in its exact support-authorized output set. This prediction shall not create rows outside that set or be promoted to compact-source, extended-source, integrated-flux, absolute-offset, or astrometric fidelity.

**SCI-MAP-REQ-015 – Projection-specific estimand.** The authorized base projection is exactly `SCI-MAP:one_hot_containing_pixel@1`; it preserves unity over the finite in-grid domain and has explicit half-open boundary loss. Fractional projection, diagonal weighted least squares, and conservation across a finite boundary are outside this branch until separately authorized.

**SCI-MAP-REQ-016 – One exact MAP operator for linear companions.** For one selected immutable realized plan  $\Pi$ ,  $A_{\text{MAP},\Pi} \equiv A_{\text{out}} \equiv D_{Q_{\text{out}}}^{-1} J_{\text{out}} G \Omega$  is the only MAP operator and its complete realized identity is the MAP-local operator-response disclosure. Its identity binds the exact admitted PTC-sample input domain, support-authorized output-row domain, coefficient family/generation, one-hot projection plan, support policy, target WCS/grid, immutable parents, and lifecycle generation. Signal, fixed-state response, conditional covariance, an available realized PTC-grid response companion, and fixed noise realizations shall use that same operator, membership, placement, coefficient state, and output rows. A PTC full-procedure difference with MAP state fixed may use the same operator only after its baseline and perturbed products, common PTC-sample comparison domain, exact input maps, common output-row comparison domain, and exact output maps are authorized; equality to  $A_{\text{MAP},\Pi} \Delta \mathbf{z}$  requires those exact common domains rather than array shape. A PTC-grid companion begins at MAP and receives the operator exactly once, never  $H_{\text{fixed},\Theta}$  again. Physical original-footprint exposure instead uses unique originals and each original's own AST ALIGN-grid coordinate; causal influence is not a linear exposure companion.

**SCI-MAP-REQ-017 – Realized response claim.** A stored kernel may be called a realized fixed-state response to template  $\tau$  only when its exact sample parent satisfies  $\mathbf{k} = H_{\text{fixed},\Theta} \tau$  and the support-restricted map satisfies  $\hat{\mathbf{k}}_{\text{out}} = A_{\text{MAP},\Pi} H_{\text{fixed},\Theta} \tau = R_{\text{fixed},\Theta} \tau$  on exactly the signal rows. A PTC full-procedure finite difference  $\Delta \mathbf{z}_{\text{PTC-FP}}$  and either map difference shall retain its distinct family identity and the complete comparison-space record, including both product identities, quantity/unit, rows, frames/WCS, reference/gauge/null state, supports, bases, common domain and exact two maps, unmatched-row treatment, perturbation and finite-difference convention, state transition, availability, and cause.  $\Delta \mathcal{S}_{\text{PTC-FP}}$  records state change but never authorizes numerical subtraction. Otherwise the response is a tracer, separately typed discontinuity, limited, or unavailable.

**SCI-MAP-REQ-018 – Absolute-offset mode.** SCI-MAP v0.2 shall apply no signal-centering operation:  $L = I$ . Coaddition shall not subtract a mean, remove a constant mode, or recenter a source; any upstream loss of that mode shall remain explicit in response and provenance.

**SCI-MAP-REQ-019 – Conditional covariance.** When a declared PTC-domain covariance  $N_{\text{PTC}}$  is available for the exact fixed membership and state  $\Theta$ , MAP shall propagate it as  $C_{m,\text{out}} = A_{\text{MAP},\Pi} N_{\text{PTC}} A_{\text{MAP},\Pi}^T$ , retaining

consequential off-diagonal terms on the exact operator domains. The equation is complete only on the declared available domain. Missing, limited, symbolic, summarized, lineage-resolvable, or unavailable terms remain typed and are never zero-filled or interpreted as independence; absence of a complete model does not invalidate the signal map.

**SCI-MAP-REQ-020 – When normalization is precision.** Normalization may be called marginal inverse variance only when the exact selected  $\gamma$  family is calibrated true inverse variance, selected errors are independent on the asserted domain, and  $\sum_i \gamma_i (G_{pi}^2 - G_{pi}) = 0$ . One-hot geometry alone does not prove the coefficient premise.

**SCI-MAP-REQ-021 – Formal diagonal weight.** When its assumptions hold,  $\phi_p = Q_p^2 / \sum_i G_{pi}^2 \gamma_i$  is the formal marginal inverse variance. It remains distinct from  $Q_p$ , support-restricted precision  $C_{m,out}^+$ , exposure, hits, and empirical statistical weight.

**SCI-MAP-REQ-022 – Covariance and nuisance disclosure.** The bundle shall report what uncertainty or covariance information it actually provides, its meaning, domain, assumptions, and limitations, and whether fuller information is persisted, lineage-resolvable, summarized, incomplete, or unavailable. It shall list omitted correlations and nuisance terms; absence of a complete covariance model shall not invalidate the map or prohibit later analysis, and unavailable information shall not be set to zero. A later covariance estimate may be attached only as a new versioned product bound to the original MAP product and processing identity.

**SCI-MAP-REQ-023 – Data-derived state.** If membership, coefficients, response, or geometry are estimated or selected from the same data, the displayed linear expectation and covariance shall remain explicitly conditional; unconditional bias or covariance requires a named joint model or resampling result outside this fixed-state claim.

**SCI-MAP-REQ-024 – No significance promotion.** Signal multiplied by the square root of merely formal weight shall be labeled formal standardized signal, not `sig2noise` or empirical significance. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps  $Q$  normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims. In particular, MAP shall not equate this quantity with NOI's conditional marginal second moment or standardized signal, and no future calibrated significance product is inferred.

**SCI-MAP-REQ-025 – Distinct map facts.** Every base/unfiltered MAP bundle shall separately represent AST-geometric incidence, MAP-route-candidate incidence, estimator-admitted contribution, admitted-observation count, `upstream_eligible_original_footprint_exposure`, `retained_original_footprint_exposure`, numerical-normalization support, science-policy support, and MAP-local base-product scientific validity. No fact or downstream eligibility decision shall become authority for another.

**SCI-MAP-REQ-026 – Incidence layers.** AST-geometric incidence records valid coordinate placement independent of signal availability; MAP-route-candidate incidence additionally requires the exact realized PTC route and occurrence-level upstream admission; estimator contribution additionally requires all remaining gates in  $C_p$ . The one-hot touch convention counts the unique containing pixel only.

**SCI-MAP-REQ-027 – Estimator contributions.** Estimator-admitted contributions shall count exactly  $i \in C_p$ . A candidate ineligible under the exact MAP admission or contribution policy for its preserved cause, or with unavailable/conflicting required state, zero/negative/non-finite coefficient, non-finite signal, or out-of-grid projection, contributes nowhere; no producer flag implies a universal action.

**SCI-MAP-REQ-028 – Admitted observation count.** For coadds, admitted observation count shall count complete observation bundles contributing at the pixel and shall not be confused with sample hits or observation attempts.

**SCI-MAP-REQ-029 – Upstream-eligible original-footprint exposure.** MAP `upstream_eligible_original_footprint_exposure` shall union and deduplicate exact positive- $e_{vo}$  original occurrences that are ancestors of route candidates admitted by `SCI-MAP:map_upstream_admission@2`. Each original is placed once by its own exact AST coordinate and the one-hot half-open rule; it is not placed at every descendant coordinate. The exposure population precedes coefficient positivity and MAP-local numerical rejection. Missing original-coordinate or lineage authority makes this exposure role unavailable.

**SCI-MAP-REQ-030 – Retained original-footprint exposure.** MAP `retained_original_footprint_exposure` shall use the same original-occurrence key and own-coordinate placement as REQ-029, restricted to originals that are ancestors of estimator-admitted members. Overlapping filters, decimation, replacement, synthesis, and multiple descendants do not duplicate seconds. Coadded exposure shall union observation-scoped original identities after atomic coadd admission; one original can occupy at most one pixel under the base one-hot family, or none after boundary loss. Neither exposure is precision.

**SCI-MAP-REQ-031 – Adopted threshold selector.** For each declared support-policy population, sort its finite strictly positive normalization coefficients  $Q_p$ ; for count  $N > 0$ , select zero-based  $k = \lfloor ([0.75N] + N)/2 \rfloor$  and  $Q_\star = Q_{(k)}$ ; for  $N = 0$ , use the exact boundary convention  $Q_\star = 0$ , which selects no population element and never denotes a zero-sky measurement. The exact population and policy identities and each reached requested, effective, observation-resolved, applied, and realized state/value of dimensionless  $c = \text{coverage\_cut}$  shall be retained; a successful realized product contains all five, while failure or nonproduction retains only the reached prefix. Pre-application admission shall check the exact observation-resolved value. Application shall bind that value to one support-policy attempt; realization shall later bind only an actual product. The exact Boolean plan fact `coverage_cut_above_one_override_requested` shall bind request identity, authorized owner or profile, scope, each reached lifecycle stage, exact value, cause, and provenance. It has no applied/realized record before application and no realized record after failure or nonproduction. The scalar shall be finite and nonnegative. Exact zero requests no relative cut while finite strictly-positive  $Q_p$  remains mandatory. Values  $0 < c \leq 1$  are the ordinary recommended range. A value  $c > 1$  is admissible only when that independently authorized fact is true for the same resolved value and scope; the number cannot authorize itself and may yield no support-authorized output rows. That prospective case is governed by [the exact empty-support disposition](#). Negative, non-finite, missing, or unrepresentable values shall fail before support-row construction, application, or required-product mutation. No universal numerical default is established.

**SCI-MAP-REQ-032 – Separate support predicates.** Only after the effective plan and exact observation-resolved cut pass REQ-031 shall application bind  $c_{\text{applied}} = c_{\text{observation\_resolved}}$  for one support-policy attempt. Numerical normalization support shall then require finite positive  $Q_p \geq Q_\star c_{\text{applied}}/10$ , and science-policy support shall independently require finite positive  $Q_p \geq Q_\star c_{\text{applied}}$ . Both comparisons are inclusive and use the exact declared population and inherited order-statistic rule, including  $Q_\star = 0$  when  $N = 0$ . At  $c = 0$ , no relative threshold excludes a row, but no zero, negative, non-finite, or absent  $Q_p$  is restored. An authorized expert  $c > 1$  may leave the science-support set empty; [the exact empty-support disposition](#) governs that applied attempt. These thresholds are an operational relative-support policy and imply no exposure, precision, significance, response, completeness, or physical optimality. Neither support state establishes final validity or precision; no realized value is required to authorize this pre-application check.

**SCI-MAP-REQ-033 – MAP-local base-product validity.** One MAP-local base-product scientific-validity state shall require membership in the exact support-authorized row set, exact admitted PTC and AST parents, passing MAP contribution/support policy, finite normalized signal, and absence of an applicable MAP-product failure. Explicitly limited or unavailable response/covariance does not alone invalidate the map. An empty support-authorized row set realizes no MAP product and therefore has no base-product validity state; [the exact empty-support disposition](#) governs it. Retained storage, finite payload, coefficient, exposure, incidence, normalization, or downstream-local eligibility cannot establish or rewrite MAP validity.

**SCI-MAP-REQ-034 – Classify before arithmetic.** Only structurally admitted payloads shall be retrieved and safely classified for finiteness/sign before membership construction. All required gates shall pass before payload arithmetic. Multiplication by zero, masking after accumulation, or sanitizing a non-finite aggregate shall not substitute for admission.

**SCI-MAP-REQ-035 – Exceptional contribution states.** Zero coefficient shall be noncontributing, a low positive coefficient shall follow the explicit adopted policy, negative or non-finite ordinary coefficients shall be invalid, and an out-of-bounds projection shall contribute nowhere. Each cause shall remain distinguishable.

**SCI-MAP-REQ-036 – Complete observation bundle and roles.** The six independently requestable product roles are `SCI-MAP:base_observation_map@1`, `SCI-MAP:response_bearing_observation_map@1`, `SCI-MAP:covariance_qualified_observation_map@1`, `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1`. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance. A `SCI-MAP:base_observation_map@1` shall exist as one immutable

ordered logical bundle before coaddition only when actual support-authorized scientific rows and product values were realized. It contains signal, numerator, normalization identity, complete MAP-local realized operator identity, honest upstream-response and uncertainty/covariance identity and status, the distinct facts in REQ-025, exposure lineage/accounting identity, unit and exact nominal-beam parent, complete AST frame/WCS, shape/indexing, estimator/projection/profile identities, PTC removed-component/additive-reference/null-space state, all five MAP lifecycle stages, successful attempt outcome, exact PTC/AST parentage, and product identity. It may remain valid when a stronger numerical upstream response or complete covariance is unavailable; those permitted companion statuses are not attempt failure. Independently requested `SCI-MAP:response_bearing_observation_map@1` and `SCI-MAP:covariance_qualified_observation_map@1` products shall each parent the exact realized base observation, bind their own lifecycle and attempt outcome, and require respectively every named exact response-family object or every named within-observation covariance block. An applied no-support attempt creates no logical observation bundle. Physical persistence is separately controlled by the effective output plan.

**SCI-MAP-REQ-037 – Role-qualified atomic coadd admission.** After exact owner acceptance, canonical Registry/source binding, and activation make `SCI-MAP:observation_coadd_admission@2` evaluable, a complete realized observation bundle shall pass every compatibility check and only the companion checks required by the exact selected product role before any coadd-owned state changes. This r0.4 candidate has no active profile or object-specific decision. Under `SCI-MAP:base_coadd@1`, unavailable or basis-incompatible response and incomplete covariance do not alter signal membership and do not reject an otherwise compatible base bundle; their exact identities, states, and causes remain. A required companion incompatibility is atomically decisive only for `SCI-MAP:response_bearing_coadd@1` or `SCI-MAP:covariance_qualified_coadd@1` as applicable. If an exact base coadd parent already exists, such an incompatibility blocks the stronger role for that parent and never selects a subset. A separately requested stronger-role admission may reject an observation only before constructing its own explicitly identified base/signal parent. Every rejection leaves all affected numerators, normalizations, counts, exposure, response, validity, provenance, and product cardinality unchanged.

**SCI-MAP-REQ-038 – Centered integer common grid.** A canonical target grid may be requested and made immutable before observation-map construction. Observation maps constructed on that exact grid, or otherwise compatible observation/coadd shapes, shall use exact centered-integer embedding; shape differences are nonnegative and even in each axis, and the corresponding integer offset maps reference pixels to the same world coordinate. A later crop is a newly identified row-selected product with immutable parent and lineage; it shall not relabel the parent or make it retrospectively compatible. Fractional shift, post-hoc pad, reprojection, interpolation, wrapping, implicit recentering, mosaicking, and a different sky grid are outside this estimator.

**SCI-MAP-REQ-039 – Ordinary coadd estimator.** The exact v0.2 family `SCI-MAP:uniform_observation_coadd_coefficient@1`, owned by MAP, assigns dimensionless  $u_{op} = 1$  to every row of each bundle eligible and realized under an exact owner-accepted, canonically registered/source-bound, active `SCI-MAP:observation_coadd_admission@2` generation and realized object decision. [the exact candidate profile-status rule](#) applies to this generation. The VAL request/applicability/eligibility/decision-artifact-realization fields remain distinct from the five-stage MAP coadd lifecycle and attempt outcome. The coefficient is an equal-observation coefficient, not precision or inverse variance. Each of `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1` is independently requestable, has its own exact base parent, lifecycle, and attempt outcome, and uses identical centered-integer equal-observation signal arithmetic for its admitted ordered membership. On exact row identities,  $N_p^c = \sum_o u_{op} \hat{m}_{op}$ ,  $Q_p^c = \sum_o u_{op}$ , and  $\hat{s}_p = N_p^c / Q_p^c$  is published only on  $\mathcal{S}_{out}^c$ . If  $\mathcal{S}_{out}^c = \emptyset$ , no coadd is realized and [the exact empty-support disposition](#) applies. Conforming in-memory and persisted bundles have identical admission, identities, arithmetic, and result; storage route is not a scientific weighting or eligibility fact.

**SCI-MAP-REQ-040 – Family-qualified coadd response.** The stacked signal vector  $\hat{\mathbf{m}}_{obs}$  shall carry explicit observation-row identities and  $B_{out}$  is the exact centered-integer equal-observation averaging operator. For one exact response family  $R$ ,  $T_{obs}^{(R)}$  is the ordered stack of the exact compatible member responses  $R_o^{(R)}$ , and  $R_{coadd,out}^{(R)} = B_{out} T_{obs}^{(R)}$  only when every member has that same family identity; source domain, perturbation definition, basis, unit, normalization, reference/gauge/null state, WCS/grid and row maps are compatible; coadd membership, centered-integer placement,  $B_{out}$ , support and coefficient state are fixed for the query; and the family is mathematically composable through that fixed linear coadd operator. `SCI-MAP:response_bearing_coadd@1` binds the exact response-family identity it carries. Fixed-state, PTC full-procedure finite-difference, PTC+MAP re-resolved, coadd-re-resolved/full-procedure, and whole-chain responses are never one unqualified family. A finite difference is not multiplied through  $B_{out}$  as though it were a linear Jacobian without an exact composition the-

orem. A perturbation that may alter observation admission, coadd membership, support, placement, coefficients, role, or plan requires a separate coadd-full-procedure or coadd-re-resolved family and its own `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. Both that coadd-procedure response and the whole-chain response are unavailable until separately authorized. `SCI-MAP:base_coadd@1` requires every admitted member's complete base bundle and all response identities/statuses, but permits unavailable or basis-incompatible response with unchanged numerical signal membership. It shall then disclose coadd response unavailable with exact cause, never a hidden response subset, zero companion, or assumed compatibility. `SCI-MAP:response_bearing_coadd@1` parents one exact base coadd with identical ordered membership and requires the exact family identity and every member  $R_o^{(R)}$  plus all fixed-state and composition conditions above. If the requirement fails for an existing parent, the stronger role is unavailable and the parent is unchanged. Any separately selected response-bearing admission that rejects a member shall construct a separately identified base/signal parent and reject before any of its state mutates. Exposure and count follow that selected admission, with exposure's unique-original placement.

**SCI-MAP-REQ-041 – Coadd covariance and precision.**  $\Sigma^{\text{obs}}$  shall be the block covariance matrix on exact stacked observation-row identities with every available within- and cross-observation block explicitly located. `SCI-MAP:base_coadd@1` requires covariance identities/statuses and permits partial, symbolic, summarized, lineage-resolvable, or unavailable state. `SCI-MAP:covariance_qualified_coadd@1` parents one exact base coadd with identical ordered membership and requires every named within- and cross-observation block before MAP may publish  $C_{c,\text{out}} = B_{\text{out}} \Sigma^{\text{obs}} B_{\text{out}}^T$ . Missing blocks are unknown and block only this stronger role; they are never zero or independence. If a block is absent for an existing parent, no subset is selected and the base coadd remains unchanged. Any separately selected covariance-qualified admission that rejects a member shall construct its own separately identified base/signal parent before mutation. The uniform observation coefficient is not precision.

**SCI-MAP-REQ-042 – Correlated-GLS boundary.** A correlated minimum-variance solution using  $(\Sigma_p^{\text{obs}})^{-1} \mathbf{1}$  may require negative combination coefficients and therefore belongs to a separately authorized correlated-GLS branch, not the ordinary positive-coefficient coadd. The token  $K_p$  remains exclusively the MAP response/kernel accumulator and never names a covariance matrix.

**SCI-MAP-REQ-043 – Minimum product and attempt identity.** Every realized observation and coadd product shall identify one exact product-role token from the six-role Registry; its independent request and all five MAP lifecycle stages; successful attempt outcome; exact base parent and ordered membership; external observation/coadd, array/group, occurrence, and row domains; estimator and coefficient families; MAP-local operator and upstream-conditioned response/covariance states; exact response-family identity for every response-bearing product; signal unit and fixed-nominal-beam parent; complete AST frame/WCS and WCS-generation identity; preconstruction target-grid request where applicable; grid/projection/boundary; exact active MAP profiles and object decisions; exact PTC product/application and coefficient generations; exposure convention; logical-bundle and plan-controlled persistence state; parentage; causes; and product identity. Any numerical full-procedure response additionally binds the complete `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record; coverage policy binds its typed above-one override record. A failed or not-produced attempt instead binds only the reached lifecycle prefix, attempt identity/outcome, exact cause, affected scope, provenance, and nonmutation evidence and has no product identity or completion marker. No STOKES token supplies quantity authority. Scientific identity shall be durable without prescribing a persistence format.

**SCI-MAP-REQ-044 – Frames and units.** MAP shall bind the complete frame identity delivered by `SCI-AST:rtc_output_grid_coordinates@1`: exact source/target frame, reference system (including ICRS or FK5/J2000 as actually declared), equinox/epoch, time scale, site/model state, TAN topology, WCS, axis order, signs, handedness, and degree-valued science axes. MAP shall not equate J2000, FK5, ICRS, apparent, observed, or unrefracted frames by name or default. Signal/response units are `mJy/beam` with exact nominal-beam identity; covariance has its square.

**SCI-MAP-REQ-045 – WCS and index persistence.** Full-precision typed or sidecar WCS shall be the lossless admission and provenance authority. For an immutable finite MAP grid of  $n_x$  by  $n_y$  cells, the 0.1-arcsec representation-test population is the finite union of (a) every scientifically addressable integer pixel center  $(j+1, k+1)$  for  $0 \leq j < n_x$ ,  $0 \leq k < n_y$ , and (b) every outer-footprint cell vertex:  $(h + \frac{1}{2}, \frac{1}{2})$ ,  $(h + \frac{1}{2}, n_y + \frac{1}{2})$  for  $0 \leq h \leq n_x$ , together with  $(\frac{1}{2}, l + \frac{1}{2})$  and  $(n_x + \frac{1}{2}, l + \frac{1}{2})$  for  $0 \leq l \leq n_y$ . The second set includes all four outer corners and every required outer-boundary vertex. The complete population is preregistered from the full-precision typed WCS, grid shape, and row domain before the physical FITS WCS is generated. Both representations shall bind the same WCS-generation identity, celestial frame, axis order, signs, handedness, orientation, shape, and centered-integer reference-pixel relation. At every population member, each WCS is evaluated at the exactly corresponding cen-

ter (zero-based memory index  $(j, k)$ , one-based FITS pixel  $(j + 1, k + 1)$ ). Separation is the great-circle angle  $\text{atan2}(\|\mathbf{u} \times \mathbf{v}\|, \mathbf{u} \cdot \mathbf{v})$  between unit direction vectors in that exact declared frame, evaluated with IEEE-754 binary64 round-to-nearest, ties-to-even operations and reported in arcseconds. The maximum over the complete finite population shall be at most 0.1 arcsec. If either WCS yields an invalid or non-finite direction at any preregistered member, the representation fails; the point is not dropped. A center is absent only when its scientific row was already absent before population generation; every listed outer-boundary vertex remains required. No smaller sampled subset or continuum/global-sky claim is authorized. The 0.1-arcsec bound is only the maximum over that exact preregistered finite population; it is not a continuum or global-sky maximum. Exact axis sign, handedness, orientation, and centered-integer shape/reference-pixel relations remain required independently of the numerical bound.

**SCI-MAP-REQ-046 – Lifecycle-bound provenance.** Every MAP plan, coefficient selection/application, projection, support policy, output plan, coadd plan, response/covariance request, product role, and coverage policy shall record each reached requested, effective, observation-resolved, applied, and realized identity in order and shall separately record application-attempt outcome. The record shall include each reached exact `coverage_cut` stage, with all five included only for successful product realization; the typed `coverage_cut_above_one_override_requested` identity, authority/profile, scope, value, cause, provenance, and binding to every reached stage; exact PTC/AST/boundary/profile and admitted coefficient Registry/family/handoff generations; the six profile semantics/approval/registration/source/activation/object-decision axes; every typed gate's independent request/applicability/eligibility/decision-artifact-realization fields plus Boolean projection, immutable reason, cause, failure scope, and DAG stage; exact product-role and base-parent identities; admitted contributions/bundles; the complete MAP-local operator  $A_{\text{MAP}, \Pi}$ , accumulators, and row domains when realized; unique-original exposure facts; PTC additive-reference/removed/null states; exact response family, composition authority, and SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE or state-transition record as applicable; covariance state; WCS comparison population/generation; centered-integer offsets; persistence state; required products; failures; and not-produced causes. No applied or realized record shall be fabricated before application. A terminal failure retains the applied attempt and failure evidence without a realized product; an applied no-support outcome retains its exact proof and cause with `not_produced` outcome and no failure or realized product.

**SCI-MAP-REQ-047 – Failure scope, nonproduction, and atomicity.** An invalid required input, unauthorized above-one cut, incompatible bundle, unrepresentable finite aggregate, unrepresentable coordinate/index conversion, missing companion required by the exact selected role, or failed required publication shall propagate at its declared contribution, pixel, observation-admission, or complete-product scope and shall fail before the affected live state changes. A pre-application failure creates no applied or realized record. A failure after application retains the exact applied attempt, failure stage, cause, affected scope, and nonmutation evidence without fabricating a realized product or completion marker. A valid applied policy yielding no support-authorized output rows is instead `not_produced` with cause `no_support_authorized_output_rows`; it is not failure and publishes no map, coadd member, response, covariance, or completion marker. Base-role response/covariance unavailability permitted by REQ-037–041 is a realized companion-status fact, not producer or attempt failure.

**SCI-MAP-REQ-048 – Required observation products.** A complete realized logical `SCI-MAP:base_observation_map@1` bundle with at least one support-authorized scientific row shall exist before coaddition. For an applied observation attempt with no such rows, the exact empty-support disposition applies, and the attempt is not an admitted coadd member. Persisted per-observation products are the normal scientific expectation and shall be published whenever the effective v0.2 output plan designates them required; every required publication shall complete or the declared operation fails without a completion claim. A plan may instead permit a fully in-memory route. A coadd-only entry may consume persisted conforming observation bundles without rerunning upstream processing. In-memory and persisted routes shall use identical logical admission, arithmetic, role and base-parent identity, lifecycle, attempt outcome, and provenance. Optional nonpublication of a realized logical product does not imply that the product did not exist; no-support nonproduction means that no logical observation product exists. This MAP rule does not waive PTC's upstream atomic required-publication obligations.

**SCI-MAP-REQ-049 – Immutable base parent.** Every stronger observation/coadd role and downstream transformation shall preserve the exact resolvable base/unfiltered MAP-bundle or base-coadd identity, ordered membership, lifecycle, and MAP-local base-product validity. A later local validity/eligibility decision may differ for its named use but shall not rewrite MAP validity, alter the original claims, sever parentage, or select a hidden subset of an existing base parent. Later response or covariance estimates are new versioned products.

**SCI-MAP-REQ-050 – Complete consumer boundary.** No consumer shall receive a bare signal plane as a scientifically complete map. The boundary shall include exact product role, request, base parent and mem-

bership; quantity and nominal-beam identity; estimator/normalization/coefficient/profile identity; honest response class/state and uncertainty/covariance disclosure; distinct map and exposure facts; PTC additive-reference/removed-component/null-space state; exact frame/WCS, shape/indexing, all reached lifecycle stages, parentage, dependency versions, comparison-space record when applicable, and failures. Later analyses choose their own exact claim requirements and may attach honest versioned response/covariance products; MAP does not own their eligibility. SCI-NOI owns its exact realization-generation, empirical-uncertainty, conditional-scale, and standardization methods and products. Any physical- noise, total-uncertainty, covariance-completeness, significance, probability, false-rate, completeness, purity, or catalogue interpretation requires a separately approved exact method and remains unavailable by default. MAP therefore keeps  $Q$  normalization, formal conditional precision, complete conditional covariance, the NOI `conditional_detector_sign_randomization_marginal_second_moment`, the NOI standardized signal, and any future calibrated-significance method as distinct identities and claims.

**SCI-MAP-REQ-051 – Same fixed noise operator.** An admitted noise realization may be described as using the same map/coadd operator only when eligibility, projection, grouping, coefficients, normalization, exact support-authorized row selection, boundary, and atomic coadd admission are fixed or identically prescribed. Re-estimation from the randomized data defines a different named estimator.

**SCI-MAP-REQ-052 – Method and claim separation.** Pointing and OOF may be registered consumers of the exact ordinary MAP gridding operator only when their inputs satisfy this operator contract. Reuse grants no mode-specific WCS/frame, source model, fitting, response interpretation, output, default, or qualification authority; an unsupplied mode profile remains unavailable. This authority shall not otherwise be applied by analogy to JINC, maximum-likelihood mapmaking, OOF residual transfer, reprojection, filtering, empirical noise, fitting, Beammap inference, feedback, Stokes Q/U, measured-R mapmaking, or other units. Independent inspection, modification, experimentation, and proposal remain unrestricted, while an altered scientific identity or official claim requires an explicit versioned successor and evidence appropriate to that claim. Algebraic contract correctness, engineering conformance, representation/response fidelity, observational performance, and production readiness shall remain separate claims.

## D Falsifiable predictions

These predictions are preregistered contract consequences or conformance challenges. Their presence is not validation evidence.

**SCI-MAP-PRED-001 – Uniform constant.** For fixed membership, a constant admitted input with uniform positive coefficients produces that constant on every support-authorized MAP-valid output row; numerator and normalization scale together, and no unsupported scientific row is created. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-002 – Unequally weighted constant.** For fixed membership, unequal positive coefficients still preserve the same constant on every support-authorized output row, while normalization and conditional variance generally change with position; rows outside that set have no normalized scientific value. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-003 – One-pixel hand calculation.** For an authorized row with admitted  $z_i$  and  $a_i > 0$ , the exact triplet is  $(N, Q, \hat{m}) = (\sum a_i z_i, \sum a_i, \sum a_i z_i / \sum a_i)$ ; an unauthorized row has no normalized scientific value. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-004 – One-hot boundary.** A finite coordinate inside exactly one half-open cell has one  $G_{pi} = 1$ , preserves unity over pixels, and contributes only there. A coordinate on an internal upper edge belongs to the adjacent cell; a coordinate on the outer upper extent or outside the grid contributes nowhere. Fractional splitting is not authorized. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-005 – Delta input.** For a declared fixed state, a basis-pixel sky perturbation produces the corresponding support-row-restricted column of  $R_{\text{fixed}, \Theta} = A_{\text{MAP}, \Pi} H_{\text{fixed}, \Theta}$ , including projection and boundary truncation. A PTC full-procedure or PTC+MAP re-resolved numerical difference exists only when its complete authorized comparison-space record supplies exact common domains and both row maps; absent that record it is unavailable or only a typed state transition/discontinuity and need not equal the fixed-state column. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-006 – Point-source template.** A declared point-source perturbation  $\tau_b$  produces  $R_{\text{fixed}, \Theta} \tau_b$  only for the exact fixed response state and authorized rows. PTC full-procedure and PTC+MAP re-resolved families additionally retain baseline/perturbed identities, perturbation side/sign/amplitude/location/source state/normalization, finite-difference convention, gauge/support/bases, exact common domain and row maps, state transition, availability, and cause. The unavailable whole-chain RTC-to-CAL-to-PTC-to-MAP family is not inferred. Peak, integrated, and fitted amplitudes remain distinct functionals. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-007 – Resolved and Fourier inputs.** At fixed state, resolved disks, gradients, and Fourier modes produce their explicit  $R_{\text{fixed}, \Theta} \tau$  on the authorized row set; a constant-input pass does not predict preservation of these modes, and procedure-response families are not inferred from the fixed matrix. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-008 – Variable coverage.** Fixed-coefficient constant preservation can coexist with spatially varying normalization, support-row membership, covariance, and nonconstant-template response; a row that fails support is absent rather than measured zero. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-009 – Finite boundary.** At an admitted finite edge, no contribution wraps or clamps; local normalization may preserve a constant while compact-source peak or integrated response changes. *Primary claim layer: Representation fidelity.*

**SCI-MAP-PRED-010 – Coefficient states.** The admitted uniform family yields exact dimensionless  $\gamma_i = 1$  only for an explicit selection with complete PTC publication, exact target/payload/occurrence association, MAP permission, and a requested, applicable, eligible, realized named handoff under the exact compatible source binding. Registry presence alone realizes nothing. Missing selection or handoff remains unavailable with no fallback. Structural availability is independent of numerical class; MAP alone classifies finite positive, exact zero, negative finite, non-finite, and unrepresentable. Zero is noncontributing and the other nonpositive/non-finite classes are invalid with exact cause; uniform one is not precision. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-011 – Use-qualified exclusion and non-finite states.** A candidate whose typed gate record projects to Boolean nonmembership does not contaminate an accumulator. Structural exclusion prevents payload

retrieval; structurally admitted payloads are read only for safe MAP numerical classification before membership and arithmetic. Its complete named-field or other typed record, immutable reason, and cause remain preserved; no universal producer-flag action is inferred. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-012 – Threshold selector.** For each predeclared positive finite coefficient population, the selected zero-based index is exactly  $\lfloor ([0.75N] + N)/2 \rfloor$ ; empty input uses the exact boundary convention  $Q_\star = 0$  while selecting no population element and asserting no zero-sky measurement. No row gains support without its own finite positive  $Q_p$ . Small  $N$  fixtures straddle every integer-index transition. At exact  $c = 0$ , both relative thresholds are zero and the finite-positive requirement still excludes zero, negative, non-finite, or absent normalization. Values  $0 < c \leq 1$  use inclusive comparisons in the ordinary recommended range. A  $c > 1$  value is admitted only with the independently true, same-value/scope `coverage_cut_above_one_override_requested` fact and may yield empty support; the number does not authorize itself. That prospective applied case has state `applied_no_support_authorized_output_rows`, outcome `not_produced`, exact cause `no_support_authorized_output_rows`, no failure, and no realized MAP product. Negative, non-finite, missing, or unrepresentable  $c$  fails before application or required-product mutation. Every reached cut stage is retained, with all five only for successful product realization; pre-application validation checks the observation-resolved value and never requires a realized value. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-013 – Missing response.** Every `SCI-MAP:base_observation_map@1` discloses its known MAP-local realized operator response on exact signal rows. Missing upstream response yields a separately typed limited or unavailable state on complete signal membership, never a smaller subset, zero, or identity. `SCI-MAP:response_bearing_observation_map@1` is unavailable for that exact base parent only when its exact selected role request names that exact response-family object as required; absence of a different fixed-state, full-procedure, re-resolved, coadd-procedure, or whole-chain family does not block an otherwise complete named response role. Each numerical full-procedure difference requires its complete `SCI-MAP-FULL-PROCEDURE-COMPARISON-SPACE` record. Coadd-full-procedure/re-resolved and whole-chain responses remain separately unavailable unless authorized. The base bundle remains usable for claims that do not require the absent response; any later response estimate or corrected map has a new versioned identity bound to the original. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-014 – Role-qualified atomic rejection.** After exact activation makes the proposed aggregate profile evaluable, a required companion incompatibility rejects the complete observation before coadd mutation at the selected role scope. Under `SCI-MAP:base_coadd@1`, unavailable or basis-incompatible response does not change signal membership; it yields unavailable coadd response with exact cause and no hidden subset or zero companion. Under `SCI-MAP:response_bearing_coadd@1`, a mixed or incompatible response-family identity, missing fixed-state condition, or absent exact composition theorem blocks that role for an existing base parent. A required covariance incompatibility analogously blocks `SCI-MAP:covariance_qualified_coadd@1`. Any separately requested admission may reject the observation only before constructing its own explicitly identified base/signal parent. Other required identity, unit, WCS, shape, policy, and parent incompatibilities remain atomic at their declared role scope. The current candidate has no active profile or realized object decision. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-015 – One-observation coadd.** After exact profile activation and a realized eligible object decision, a `SCI-MAP:base_coadd@1` containing one admitted base observation under the uniform dimensionless coefficient family reproduces its signal on corresponding centered-integer authorized rows. If the valid applied coadd support policy authorizes no rows, outcome is instead `not_produced` with exact no-support cause and no zero coadd. Conforming in-memory and persisted bundles give the same admission, identity, and arithmetic. Unavailable or basis-incompatible upstream response leaves nonempty signal membership unchanged and makes coadd response unavailable with exact cause. `SCI-MAP:response_bearing_coadd@1` reproduces response only when it binds one exact family  $R$ , every  $R_o^{(R)}$  and all fixed-state/composition conditions are compatible, and  $R_{\text{coadd,out}}^{(R)} = B_{\text{out}} T_{\text{obs}}^{(R)}$  is authorized. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-016 – Uniform observation coefficients.** On each authorized coadd row, two compatible observations produce their equal-observation arithmetic mean and count-like normalization under `SCI-MAP:uniform_observation_coadd_coefficient@1`. Each requested coadd role binds its own exact parent, lifecycle, attempt outcome, and ordered membership; stronger companions do not change the arithmetic membership of their exact base parent. A response-bearing coadd additionally binds one exact compatible response family and the fixed coadd state/composition authority; a finite difference without an exact composition theorem is not propagated as a Jacobian. Coadd-full-procedure/re-resolved and whole-chain response remain unavailable. The normalization is not precision. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-017 – Missing, invalid, or no-support observation.** An unsupported observation row contributes at no coadd row; an observation rejected by the exact selected role contributes nowhere to that separately identified base/signal parent and does not increment its count. An applied observation attempt with no support-authorized output rows has outcome `not_produced`, is not an admitted coadd member, and has no signal, response, covariance, exposure, count, validity, or completion product. A companion unavailable but permitted by a realized base role does not remove the observation and is not attempt failure. No case creates a zero-valued scientific or companion substitute. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-018 – Centered integer placement.** A canonical target grid requested before construction yields observation maps already bound to that grid. Other compatible shape differences that are nonnegative and even produce the exact half-difference row/column offset; odd, negative, fractional, or WCS-inconsistent relations fail with no mutation. The FITS representation comparison evaluates every preregistered addressable pixel center and outer-boundary cell vertex, including all corners, with the exact spherical metric and invalid-point rule. A later crop is a new row-selected product with its own identity and lineage and cannot relabel its parent as compatible. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-019 – Cross-observation covariance.** For two maps with variances  $\sigma_1^2, \sigma_2^2$  and covariance  $c$ , a complete `SCI-MAP:covariance_qualified_coadd@1` over the exact base-coadd membership includes  $c$  through  $\Sigma^{\text{obs}}$ . If a required block is missing, that stronger role is unavailable for the parent; the base role retains a partial, symbolic, summarized, lineage-resolvable, or unavailable state. No subset is formed, the block is never zero or independence, and equal-observation normalization is not precision. *Primary claim layer: Algebraic contract.*

**SCI-MAP-PRED-020 – Fixed noise operator.** A deterministic admitted realization propagated with fixed state yields the same support-row identities and coefficient-by-coefficient  $A_{\text{MAP}, \Pi} \equiv A_{\text{out}}$  and  $B_{\text{out}}$  as the signal path; recomputed state exposes a named difference. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-021 – Sequential and parallel reduction.** Integer facts, identities, masks, support, and cardinality agree exactly. Floating sums agree with a preregistered operation-count/conditioning bound that is frozen before results are viewed and is not loosened after failure. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-022 – No false significance.** A deliberate attempt to label signal times square root of merely formal weight as empirical significance is rejected or remains unmistakably labeled formal standardized signal. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-023 – No validity promotion.** A downstream product may have its own use-local validity or eligibility but cannot change MAP-local base-product validity or sever the base/unfiltered MAP parent. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-024 – No ordinary predicate for JINC.** A deliberate attempt to apply positive-coefficient ordinary admission, support, or complete-bundle availability to JINC is rejected as outside SCI-MAP v0.2. *Primary claim layer: Engineering conformance.*

**SCI-MAP-PRED-025 – Quantity-specific zero, index, and no-support outcomes.** A finite exact-zero signal numerator  $N_p = 0$  with finite positive normalization  $Q_p > 0$ , valid support, and every other required gate passing is admissible and gives the valid normalized scientific value  $\hat{m}_p = 0$ . An in-bounds zero-based index and a zero centered-integer placement offset are valid when their exact domain and embedding rules pass. A row with  $Q_p = 0$  is outside finite-positive normalization support: division is unauthorized and no supported scientific row is formed. This normalization-specific result is not a generic rule that zero is invalid. A zero coefficient remains noncontributing; `coverage_cut` = 0 retains its no-relative-cut meaning; and support-population count  $N = 0$  retains  $Q_\star = 0$  while selecting no population element and authorizing no row by itself. Non-finite, overflowed, out-of-domain, and unrepresentable quantities follow their exact declared failure scopes before affected mutation or completion. Separately, a valid applied policy whose support-authorized output-row set is empty produces no product with exact `not_produced` outcome and cause `no_support_authorized_output_rows`; it is not a failure and creates no fabricated zero substitute. *Primary claim layer: Engineering conformance.*

## E Product vocabulary

The six independently requestable product roles are `SCI-MAP:base_observation_map@1`, `SCI-MAP:response_bearing_observation_map@1`, `SCI-MAP:covariance_qualified_observation_map@1`, `SCI-MAP:base_coadd@1`, `SCI-MAP:response_bearing_coadd@1`, and `SCI-MAP:covariance_qualified_coadd@1`. Every request has its own immutable identity, parent set, ordered MAP lifecycle, result, and failure scope. A stronger role is a new product over the exact base parent; it does not rewrite that parent's signal membership, response/covariance status, validity, or provenance.

| Product                                                                                      | Unit                                       | Contract meaning                                                                                                                                                                                     |
|----------------------------------------------------------------------------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Signal numerator                                                                             | coefficient times<br><code>mJy/beam</code> | Pre-normalized accumulator; not a sky estimator.                                                                                                                                                     |
| Normalization                                                                                | declared gridding<br>unit                  | Positive denominator; precision only under Requirements 020–021.                                                                                                                                     |
| Normalized signal                                                                            | <code>mJy/beam</code>                      | Conditional ordinary map estimand whose vector contains exactly the effective-policy-authorized output rows; no unsupported zero-filled semantic extension.                                          |
| Formal diagonal weight                                                                       | inverse square signal<br>unit              | Conditional marginal inverse variance under named assumptions; not empirical significance.                                                                                                           |
| Conditional covariance                                                                       | square signal unit                         | Full fixed-state noise propagation on the exact scientific output-row domain, with omissions and representation status named.                                                                        |
| Response/kernel                                                                              | <code>mJy/beam</code> when<br>unit-bearing | Realized template response only when the sample-parent identity is proven; otherwise tracer or unavailable.                                                                                          |
| Base observation/coadd<br>roles                                                              | signal unit                                | Complete base signal bundle with known MAP-local operator and honest response/covariance identities and states; unavailable stronger companions do not change signal membership.                     |
| Response-bearing roles                                                                       | response-dependent                         | Exact child of its base parent; requires only every numerical response object named by the selected role on compatible source domain, basis, class, unit, normalization, parent, WCS/grid, and rows. |
| Covariance-qualified roles                                                                   | square signal unit                         | Exact child of its base parent; requires every within- and cross-observation block named by the role in $\Sigma^{\text{obs}}$ . Missing is unknown and blocks only that role.                        |
| NOI conditional marginal<br>second moment                                                    | square signal unit                         | Exact NOI method product; not physical-noise variance, total uncertainty, complete covariance, precision, weight, or significance.                                                                   |
| NOI standardized signal                                                                      | dimensionless                              | MAP signal standardized by the stated conditional detector-sign-randomization second-moment scale; no significance or probability interpretation.                                                    |
| Geometric/estimator hits                                                                     | integer                                    | Counts of declared populations, not coverage, precision, or validity.                                                                                                                                |
| <code>upstream_eligible_</code><br><code>original_footprint_</code><br><code>exposure</code> | named<br>time/accounting<br>unit           | Unique-original acquisition-footprint accounting for route-candidate ancestors; explicit prose alias: <code>upstream-eligible exposure</code> ; not precision.                                       |
| <code>retained_original_</code><br><code>footprint_exposure</code>                           | named<br>time/accounting<br>unit           | Unique-original acquisition-footprint accounting for estimator-admitted ancestors; explicit prose alias: <code>retained exposure</code> ; not precision.                                             |
| Support facts                                                                                | logical                                    | Separate normalization and science-policy predicates; their effective conjunction defines output-row membership.                                                                                     |
| MAP-local base-product<br>validity                                                           | logical                                    | Authority for the base/unfiltered MAP bundle only; downstream uses own their eligibility.                                                                                                            |

## F Admitted source register and independence statement

The scientific author used only the owner-approved, content-bound packet:

| Source                  | Exact admitted identity                                                                                                                                                                                                                                                                          |
|-------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Scope Brief             | Reviewed exact r0.4 manager brief; SHA-256 d3794abde9cb0c4e cec9aee447df1bdbbc7250f6033350302345573a4c52aff5; the admitted original 2026-09-07 Scope Brief remains roadmap provenance.                                                                                                           |
| Author manifest         | 56 exact references; SHA-256 83baf583f73784bebc6724025fd110e5 545303f303c70af590b53c5696ee2c4f; every listed file verified.                                                                                                                                                                      |
| Owner r0.4 directive    | SCI-MAP-OWNER-DIRECTIVE-v0.2-r0.4/2026-09-08; SHA-256 34381dc7b1a19ddf6168e2259060fd808e97f8eac2b69074fbea983f02fb017b.                                                                                                                                                                          |
| MAP successor decisions | 2026-08-28 approved post-freeze record; SHA-256 4d2b857b7ec9efe4 89fe065f464df4ecd23b57a4c1320cda6a10a56592825d1c.                                                                                                                                                                               |
| Frozen MAP predecessor  | SCI-MAP v0.1/r0.7.1 freeze, source manifest, shared wrapper and six common modules; exact hashes are in the approved 56-reference manifest.                                                                                                                                                      |
| Uniform coefficient     | SCI-PTC:analysis_gridding_coefficients@draft-0.1, SCI-PTC:uniform_constant@draft-0.1, and SCI-PTC:uniform_coefficient_handoff@draft-0.1; approved common-source SHA-256 8bc42deb69e23c757883834f7cdb82b94e89210015333f48 3d15c32377a56685.                                                       |
| PTC-to-MAP boundary     | SCI-PTC_TO_SCI-MAP_BOUNDARY v0.2-draft.1/r0.4; SHA-256 58291e1c879dc911 6e81a9e3fd4323caf79a176c51e296d8119efb6937272ef8.                                                                                                                                                                        |
| MAP/AST/VAL profiles    | SCI-MAP:map_upstream_admission@2, SCI-MAP:one_hot_containing_pixel@1, SCI-MAP:observation_coadd_admission@2, SCI-MAP:uniform_observation_coadd_coefficient@1, SCI-AST_TO_SCI-MAP_ORIGINAL_FOOTPRINT_COORDINATE v0.1/r0.1, and the exact admitted uniform VAL Registry/source-binding generation. |
| Adjacent limits         | Exact approved PTC, AST, VAL, and NOI freezes/requirements enumerated in the 56-reference manifest; no adjacent science is expanded.                                                                                                                                                             |

No implementation, test, audit, repair, validation, reduction, production- status record, manager dossier, unlisted link, repository history, sibling repository, or web scientific source entered this authorship. The frozen ordinary derivation and admitted uniform family were incorporated rather than rederived. The r0.4 candidate supplies reviewable MAP and VAL source/profile binding records with exact local hashes. Their future canonical commit, owner acceptance, activation, and frozen identities remain pending and are not invented here. Acceptance of r0.3 as the architecture basis is not acceptance of these final r0.4 bytes.